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Part I

Mechanics of Mobile Robotics

Sizzling Saturn, we've got a lunatic robot on our hands.

(Isaac Asimov, I, Robot)

Chapter 1

Locomotion

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1.1 Introduction

An Autonomous Mobile Robotics (AMR) needs locomotion mechanisms which enable it to move unbounded throughout its environment. However, as with everything in engineering our solution comes with options, and so the selection of a robot's approach to locomotion is an important aspect of mobile robot design. In laboratory settings, there are robots that can walk, jump, run, slide, swim, fly and of course roll.

> Most locomotion mechanisms have been inspired by biological counterparts, shown in **Fig.** 1.1.

There is, however, one (1) exception where there is, practically, **NO** natural equivalent:

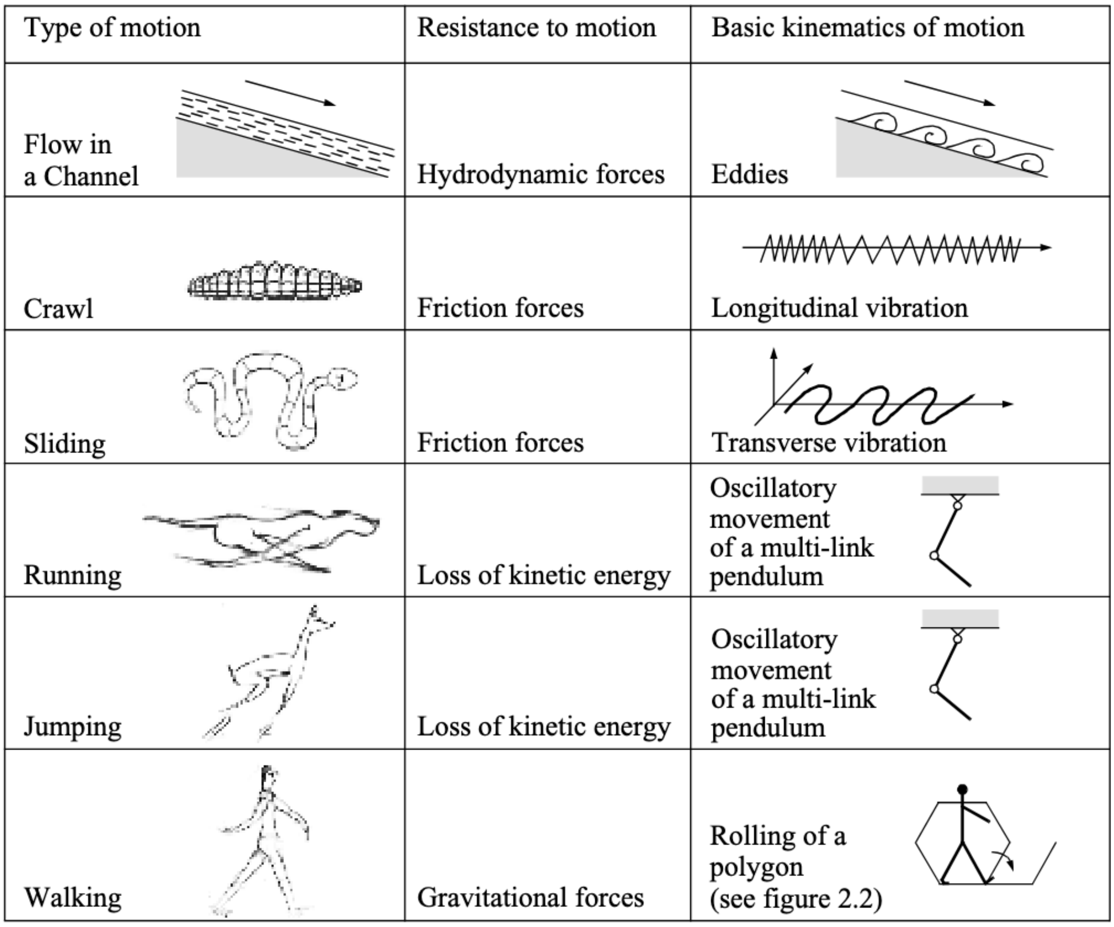
> Actively powered wheel is a human invention achieving high efficiency on flat ground.

This mechanism is **NOT** completely foreign to biological systems[1]. Our bi-pedal walking system can be approximated by a rolling polygon, with sides equal in length to the span of the step. As the step size decreases, the polygon approaches a circle or wheel. But nature did not develop a fully rotating, actively powered joint, which is the technology necessary for wheeled locomotion.

Biological systems succeed in moving through a wide variety of harsh environments. Therefore it can be desirable to copy their selection of locomotion mechanisms[2]. Replicating nature in this regard, however, is extremely difficult for several reasons.

[1] While this statement is practically true, single cell organism use a similar locomotion to what we call wheel [2].

[2] Scientifically, this is called **Biomimetics** [3]



Type of motion	Resistance to motion	Basic kinematics of motion
Flow in a Channel	Hydrodynamic forces	Eddies
Crawl	Friction forces	Longitudinal vibration
Sliding	Friction forces	Transverse vibration
Running	Loss of kinetic energy	Oscillatory movement of a multi-link pendulum
Jumping	Loss of kinetic energy	Oscillatory movement of a multi-link pendulum
Walking	Gravitational forces	Rolling of a polygon (see figure 2.2)

Figure 1.1: Types of locomotion mechanisms used in biological systems [1].

- Mechanical complexity is easily achieved in biological systems through structural replication.

  Cell division, in combination with specialisation, can readily produce a millipede with several hundred legs and several tens of thousands of individually sensed cilia[3]. In man-made structures, each part must be fabricated individually, and therefore, no such economies of scale exist.

- Cell is a microscopic building block that enables extreme miniaturisation. With very small size and weight, insects achieve a level of robustness that we have not been able to match with human fabrication techniques.

- The biological energy storage system and the muscular and hydraulic activation systems used in animals and insects achieve torque, response time and conversion efficiencies that far exceed similarly scaled man-made systems.

Based on these aforementioned limitations, mobile robots generally generate motion, either using wheeled mechanisms, a well-known human technology for vehicles, or using a small number of articulated legs, the simplest of the biological approaches to locomotion (shown in **Fig.** 1.2).

[3]short eyelash-like filament that is numerous on tissue cells of most animals and provides the means for locomotion of protozoans of the *phylum Ciliophora*

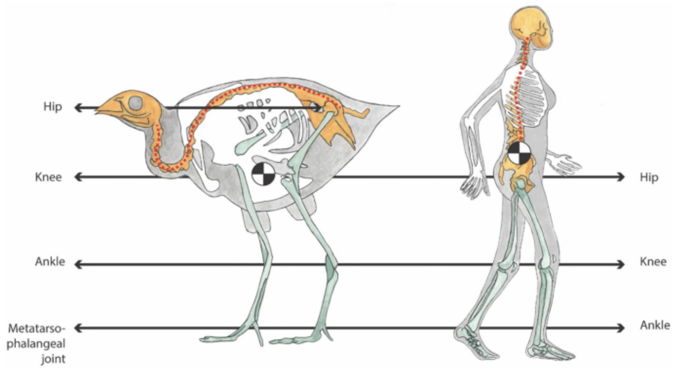


Figure 1.2: Bipedal motion is not unique to only humans as a wide variety of animals show bipedal motion [4].

In general, legged locomotion requires higher DoF and therefore greater mechanical complexity than wheeled locomotion [5]. Wheels, in addition to being simple, are extremely well suited to flat ground. As **Fig.** 1.3 depicts, on flat surfaces wheeled locomotion is one to two orders of magnitude more efficient than legged locomotion.

> The railway is ideally engineered for wheeled locomotion because rolling friction is minimised using a hard and flat steel surface.

But as the surface becomes soft, wheeled locomotion accumulates inefficiencies due to **rolling friction** while legged locomotion suffers much less because it consists only of **point contacts** with the ground. This is demonstrated in figure 2.3 by the dramatic loss of efficiency in the case of a tire on soft ground.

> the efficiency of wheeled locomotion depends on environmental qualities, such as the flatness and hardness of the ground, while the efficiency of legged locomotion depends on the leg mass and body mass, both of which the robot must support at various points in a legged gait.

It is understandable therefore nature favours legged locomotion, as locomotion systems in nature must operate on rough and unstructured terrain. For example, in the case of insects in a forest the

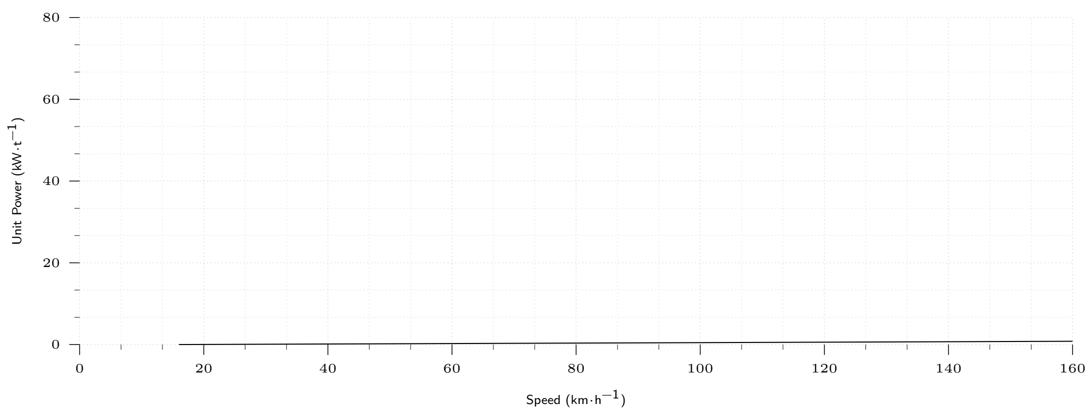


Figure 1.3: Specific power versus attainable speed of various locomotion mechanisms (Adapted from [1]).

vertical variation in ground height is often an order of magnitude greater than the total height of the insect.

By the same token, the human environment frequently consists of engineered, smooth surfaces both indoors and outdoors. Therefore, it is also understandable that virtually all industrial applications of mobile robotics utilise some form of wheeled locomotion. Recently, for more natural outdoor environments, there has been some progress toward hybrid and legged industrial robots such as the forestry robot [6] shown in **Fig.** 1.4.

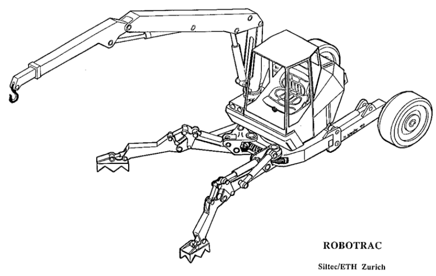


Figure 1.4: RoboTrac, a hybrid wheel-leg vehicle for rough terrain.

In the next section, we present general considerations that concern all forms of mobile robot locomotion. Following this will be overviews of legged locomotion and wheeled locomotion techniques for mobile robots.

1.1.1 Key Issues for Locomotion

Locomotion is the complement of manipulation:

- In manipulation, the robot arm is fixed but moves objects in the workspace by imparting force to them.
- In locomotion, the environment is fixed and the robot moves by imparting force to the environment.

For both cases, the scientific basis is the study of actuators which generate interaction forces, and mechanisms that implement desired kinematic and dynamic properties. Locomotion and manipulation therefore share the same core issues of stability, contact characteristics and environmental type:

- Stability
  - number and geometry of contact points
  - centre of gravity
  - static/dynamic stability
  - inclination of terrain
- characteristics of contact
  - contact point/path size and shape
  - angle of contact
  - friction
- type of environment
  - structure
  - medium (e.g., water, air, soft or hard ground)

A theoretical analysis of locomotion begins with mechanics and physics. From this starting point, we can formally define and analyse all manner of mobile robot locomotion systems. However, this

Lecture Book puts more emphasis on the mobile robot navigation problem, particularly on the topics of perception, localisation and cognition. Therefore, we will not delve deeply into the physical basis of locomotion. Nevertheless, two remaining sections in this chapter present overviews of issues in legged locomotion and wheeled locomotion.

1.2 Legged Mobile Robots

Legged locomotion is characterised by a series of point contacts between the robot and the ground. The primary advantages include adaptability and manoeuvrability in rough terrain. Because only a set of point contacts is required, the quality of the ground between those points does not matter, so long as the robot can maintain adequate ground clearance. In addition, a walking robot is capable of crossing a hole or chasm so long as its reach exceeds the width of the hole. A final advantage of legged locomotion is the potential to manipulate objects in the environment with great skill.

Figure 1.5: Dug-beetle are a great example for legged mobile robotics [7] as not only they can manoeuvre in their environment using legged motion, they can also manipulate their environment and generate rotational motion [8].

> The dung beetle, is capable of rolling a ball while locomotion as a result of its dexterous front legs shown in **Fig.** 1.5.

The main disadvantages of legged locomotion include power and mechanical complexity. The leg, which may include several DoF, must be capable of sustaining part of the robot's total weight, and in many robots must be capable of lifting and lowering the robot. Additionally, high manoeuvrability will only be achieved if the legs have a sufficient number of DoF to impart forces in a number of different directions.

Leg Configurations and Stability

Because legged robots are biologically inspired, it is instructive to examine biologically successful legged systems. A number of different leg configurations have been successful in a variety or organisms seen in **Fig.** 1.6.

Large animals such as mammals and reptiles have four (4)legs whereas insects have six (6)or more legs. In some mammals, the ability to walk on only two (2)legs has been perfected. Especially in the case of humans, balance has progressed to the point that we can even jump with one leg[4]. This

[4] In child development, one of the tests used to determine if the child is acquiring advanced locomotion skills is the ability to jump on one leg [12].

(a) Bipedal motion [9].

(b) Quadpedal motion [10].

(c) Hexapedal motion [11]

Figure 1.6: Types of motions used by different animals.

exceptional manoeuvrability comes at a price:

> Bipedal motion is much more complex active control to maintain balance.

In contrast, a creature with three (3) legs can exhibit a static, stable pose provided that it can ensure that its centre of gravity is within the tripod of ground contact. Static stability, demonstrated by a three-legged stool, means that balance is maintained with no need for motion. A small deviation from stability[5] is passively corrected towards the stable pose when the upsetting force stops. But a robot must be able to lift its legs in order to walk. To achieve static walking, a robot **must** have at least six (6)legs [13]. In such a configuration, it is possible to design a gait[6] in which a statically stable tripod of legs is in contact with the ground at all times.

[5]e.g., such as gently pushing the stool

[6]the pattern of steps of an animal at a particular speed.

Insects[7] are immediately able to walk when born. For them, the problem of balance during walking is relatively simple. Mammals, with four (4) legs, cannot achieve static walking, but are able to stand easily on four (4) legs. Fauns[8], for example, spend several minutes attempting to stand before they are able to do so, then spend several more minutes learning to walk without falling [14]. Humans, with two (2) legs, cannot even stand in one place with static stability. Infants require months to stand and walk, and even longer to learn to jump, run and stand on one leg.

[7]such as spiders, ant, beetles, . . .

[8]a baby deer

There is also the potential for great variety in the complexity of each individual leg. Once again, the biological world provides ample examples at both extremes. For instance, in the case of the caterpillar, each leg is extended using hydraulic pressure by constricting the body cavity and forcing an increase in pressure, and each leg is retracted longitudinally by relaxing the hydraulic pressure, then activating a single tensile muscle that pulls the leg in towards the body, seen in **Fig.** 1.7. Each leg has only a single DoF, which is oriented longitudinally along the leg. Forward locomotion depends on the hydraulic pressure in the body, which extends the distance between pairs of legs. The caterpillar leg is therefore mechanically very simple, using a minimal number of extrinsic muscles to achieve complex overall locomotion.

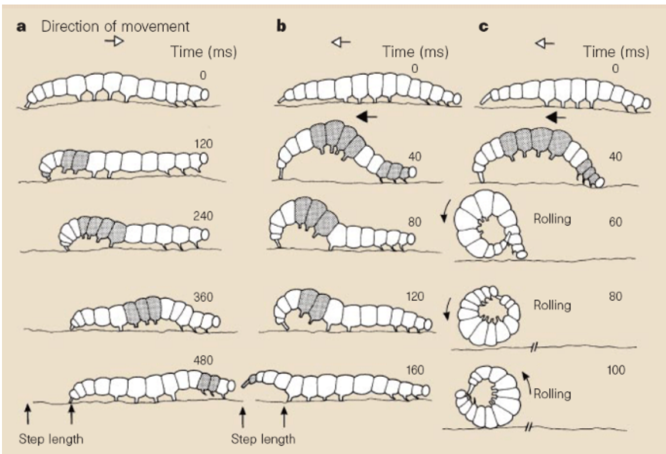


Figure 1.7: Main locomotory gaits in *Pleurotya* caterpillar [15].

At the other extreme, the human leg has more than six (6) major degrees of freedom, combined with further actuation at the toes, shown in **Fig.** 1.8. There are more than 50 muscles in each lower limb and at least half of them participate actively in the control of leg motion [16], [17].

In the case of legged mobile robots, a minimum of two (2) DoF is generally required to move a leg forward by lifting the left and swinging it forward. More common is the addition of a 3rd DoF for more complex manoeuvres, resulting in legs such as those shown in **Fig.** 1.9. Recent successes in the creation of bipedal walking robots have added a fourth DOF at the ankle joint [20]. The ankle enables more consistent ground contact by actuating the pose of the sole of the foot. In general, adding DoF to a robot leg increases the manoeuvrability of the robot, both augmenting the range of terrains on which it can travel and the ability of the robot to travel with a variety of gaits. The primary disadvantages of additional joints and actuators is, of course, energy, control and mass. Additional actuators require energy and control, and they also add to leg mass, further increasing power and load requirements on existing actuators.

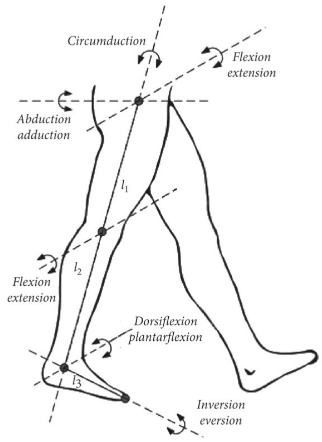


Figure 1.8: The DoF a human leg has [18].

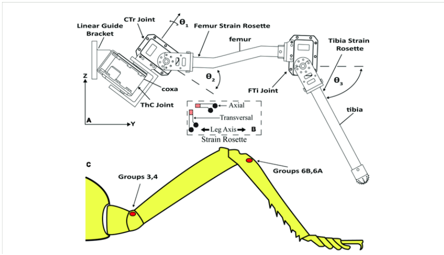


Figure 1.9: An example of a leg possessing three (3)DoF [19].

In the case of a multi-legged mobile robot, there is the issue of leg coordination for locomotion, or gait control.

> The number of possible gaits depends on the number of legs [21].

The gait is a sequence of lift and release events for the individual legs. For a mobile robot with k legs, the total number of possible events N for a walking machine is:

$$N = (2k - 1)! \tag{1.1}$$

For a bipedal walker (k=2) legs the number of possible events N is:

$$N = (2k - 1)! = 3! = 3 \cdot 2 \cdot 1 = 6 \tag{1.2}$$

The six (6) different events are:

- lift right leg
- lift left leg
- release right leg
- release left leg
- lift both legs together
- release both legs together

As can we see, this list of possible events quickly grows quite large. For example, a robot with six (6) legs has far more gaits theoretically:

$$N = 11! = 3{,}991.680.0 \times 10^{7} \tag{1.3}$$

1.2.1 Examples of Legged Robot Locomotion

Although there are no high-volume industrial applications to date, legged locomotion is an important area of long-term research. Several interesting designs are presented below, beginning with the one-legged robot and finishing with six-legged robots.

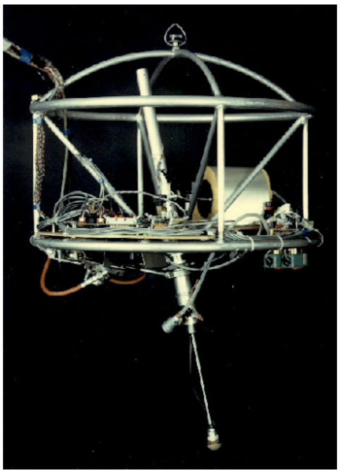


Figure 1.10: The Raibert hopper [22].

Single Leg

The minimum number of legs a legged robot can have is, of course, one. Minimising the number of legs is beneficial for several reasons.

- Body mass is particularly important to walking machines, and the single leg minimises cumulative leg mass.
- Leg coordination is required when a robot has several legs, but with one leg no such coordination is needed.
- The one-legged robot maximises the basic advantage of legged locomotion: legs have single points of contact with the ground in lieu of an entire track as with wheels.

> A single legged robot requires only a sequence of single contacts, making it useful in rough terrain.

Perhaps most importantly, a hopping robot can dynamically cross a gap that is larger than its stride by taking a running start, whereas a multi-legged walking robot that cannot run is limited to crossing gaps that are as large as its reach.

The major challenge of creating a single-leg robot is balance. For a robot with one leg, static walking is not only impossible, but static stability when stationary is also impossible. The robot must actively balance itself by either changing its centre of gravity or by imparting corrective forces. Thus, the successful single-leg robot **must be dynamically stable**.

Fig. 1.10 shows the Raibert Hopper [23], [24], one of the most well-known single-leg hopping robots created.

This robot makes continuous corrections to body attitude and to robot velocity by adjusting the leg angle with respect to the body. The actuation is hydraulic, including high-power longitudinal extension of the leg during stance to hop back into the air. Although powerful, these actuators require a large, off-board hydraulic pump to be connected to the robot at all times. **Fig.** 1.11 shows a more energy efficient design developed [26]. Instead of supplying power by means of an off-board hydraulic pump, the Bow Leg Hopper is designed to capture the kinetic energy of the robot as it lands using an efficient bow spring leg. This spring returns approximately 85% of the energy, meaning that stable hopping requires only the addition of 15% of the required energy on each hop. This robot, which is constrained along one axis by a boom, has demonstrated continuous hopping for 20 minutes using a single set of batteries carried on board the robot. As with the Raibert Hopper, the Bow Leg Hopper controls velocity by changing the angle of the leg to the body at the hip joint. The paper of Ringrose [27] demonstrates the very important duality of mechanics and controls as applied to a single leg hopping machine. Often clever mechanical design can perform the same operations as complex active control circuitry. In this robot, the physical shape of the foot is exactly the right curve so that when the robot lands without being perfectly vertical, the proper corrective force is provided from the impact, making the robot vertical by the next landing. This robot is dynamically stable, and is furthermore passive.

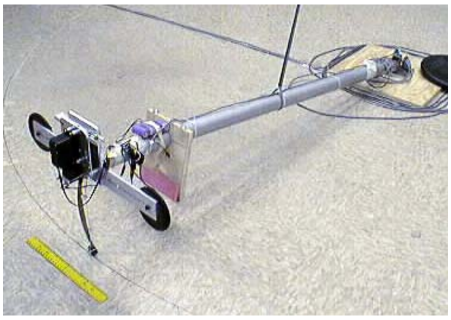


Figure 1.11: The 2D single Bow Leg Hopper.

Figure 1.12: The New ASIMO introduced in 2005 [25].

> The correction is provided by physical interactions between the robot and its environment, with no computer nor any active control in the loop.

Two Legs (Bipedal)

A variety of successful bipedal robots have been demonstrated. Two-legged robots have been shown to run, jump, travel up and down stairs and even do aerial tricks such as somersaults. **Fig.** 1.12 shows the Honda P2 bipedal robot, which is the product of tens of millions of research dollars and more than a decade of work. This biped can walk on slopes, climb and descend stairs, and push shopping carts. The crucial technology that enables this robot is Honda's research into the fabrication of extremely high torque, low mass motors that serve as the robot's joints. In the case of P2, the most significant obstacle that remains is energy capacity, efficiency and autonomous navigation. This robot can operate for only about 20 minutes with on-board power. An important feature of bipedal robots is their anthropomorphic shape. They can be built to have the same approximate dimensions as humans, and this makes them excellent vehicles for research in human-robot interaction.

Bipedal robots can only be statically stable within some limits, and so robots such as P2 and Wabian generally must perform continuous balance-correcting servoing even when standing still. Furthermore, each leg must have sufficient capacity to support the full weight of the robot. In the case of four-legged robots, the balance problem is facilitated along with the load requirements of each leg. An elegant design of a biped robot is the Spring Flamingo of MIT seen in **Fig.** 1.13. This robot inserts springs in series with the leg actuators to achieve a more elastic gait. Combined with "kneecaps" that limit knee joint angles, the Flamingo achieves surprisingly biomimetic motion.

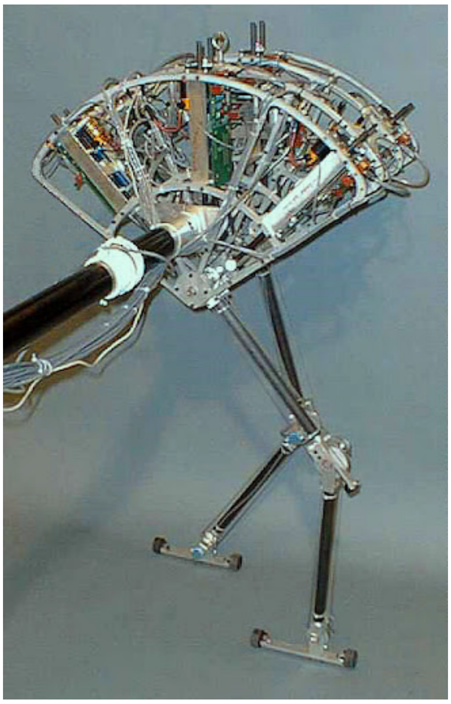


Figure 1.13: Spring Flamingo is a planar bipedal walking robot [28].

Four Legs (Quadruped)

Although standing still on four legs is passively stable, walking remains challenging because to remain stable the robot's center of gravity must be actively shifted during the gait. Sony recently invested several million dollars to develop a four-legged robot (figure 2.14). To cre- ate this robot, Sony created both a new robot operating system that is near real-time and invented new geared servomotors that are sufficiently high torque to support the robot, yet backdriveable for safety. In addition to developing custom motors and software, Sony incorporated a color vision system that enables Aibo to chase a brightly colored ball. The robot is able to function for at most one hour before requiring recharging. Early sales of the robot have been very strong, with more than 60,000 units sold in the first year. Nevertheless, the number of motors and the technology investment behind this robot dog have resulted in a very high price of approximately 1500. Four legged robots have the potential to serve as effective artifacts for research in human- robot interaction (fig. 2.15). Humans can treat the Sony robot, for example, as a pet and might develop an emotional relationship similar to that between man and dog. Furthermore, Sony has designed Aibo's walking style and general behavior to emulate learning and maturation, resulting in dynamic behavior over time that is more interesting for the owner who can track the changing behavior. As the challenges of high energy storage and motor technology are solved, it is likely that quadruped robots much more capable than Aibo will become common throughout the human environment.

Six Legs (Hexapod)

Six legged configurations have been extremely popular in mobile robotics because of their static stability during walking, thus reducing the control complexity (figure 2.16 and 2.17). In most cases, each leg has 3 DOF, including hip flexion, knee flexion and hip abduction (figure 2.6).

Figure 1.14: Genghis, one of the most famous walking robots from MIT uses hobby servomotors as its actuators.

Genghis is a commercially available hobby robot that has six legs, each of which has 2 DOF provided by hobby servos (figure 2.18). Such a robot, which consists only of hip flexion and hip abduction, has less maneuverability in rough terrain but performs quite well on flat ground. Because it consists of a straightforward arrangement of servo motors and straight legs, such robots can be readily built by a robot hobbyist. Insects, which are arguably the most successful locomoting creatures on earth, excel at traversing all forms of terrain with six legs, even upside down. Currently, the gap between the capabilities of six-legged insects and artificial six-legged robots is still quite large. Interestingly, this is not due to a lack of sufficient numbers of degrees of freedom on the robots. Rather, insects combine a small number of active degrees of freedom with passive structures, such as microscopic barbs and textured pads, that increase the gripping strength of each leg significantly. Robotic research into such passive tip structures has only recently begun. For example, a research group is attempting to recreate the complete mechanical function of the cockroach leg (Roland, reference in notes (Espenschied et al.)). It is clear from all of the above examples that legged robots have much progress to make before they are competitive with thei24 Autonomous Mobile Robots have been realised recently, primarily due to advances in motor design. Creating actuation systems that approach the efficiency of animal muscle remains far from the reach of robotics, as does energy storage with the energy densities found in organic life forms

1.3 Wheeled Mobile Robots

The wheel has been by far the most popular locomotion mechanism in mobile robotics and in man-made vehicles in general.[9] It can achieve high efficiencies, as demonstrated in figure 2.3, and does so with a relatively simple mechanical implementation. In addition, balance is not usually a research problem in wheeled robot designs, because wheeled robots are almost always designed so that all wheels are in ground contact at all times.

[9]This should be clear as wheel motion is one of the most efficient method of converting energy to motion.

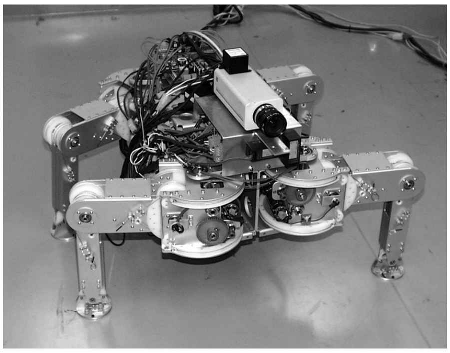


Figure 1.15: Genghis, one of the most famous walking robots from MIT uses hobby ser- vomotors as its actuators.

Therefore, three wheels are sufficient to guarantee stable balance, although as we will see below, two-wheeled robots can also be stable. When more than three wheels are used, a suspension system is required in order to allow all wheels to maintain ground contact when the robot encounters uneven terrain. Instead of worrying about balance, researchers in wheeled robots tends to focus on the problems of traction and stability, maneuverability and control:

> can the robot wheels provide sufficient traction and stability for the robot to cover all of the desired terrain, and does the robot's wheeled configuration enable sufficient control over the velocity of the robot?

1.3.1 Design

As we will see, there is a very large space of possible wheel configurations when we considers possible techniques for mobile robot locomotion. We will begin by discussing the wheel in detail, as there are a number of different wheel types with specific strengths and weaknesses. Then, we will examine complete wheel configurations that deliver particular forms of locomotion for a mobile robot.

Wheel Design

There are four (4) major wheel classes, as shown in **Fig.** 1.16. They differ widely in their kinematics, and therefore the choice of wheel type has a large effect on the overall kinematics of the mobile robot.

The standard wheel and the castor wheel have a primary axis of rotation and therefore are highly directional. To move in a different direction, the wheel must be steered first along a vertical axis. The key difference between these two (2) wheels is that the standard wheel can accomplish this steering motion with no side effects, as the centre of rotation passes through the contact patch

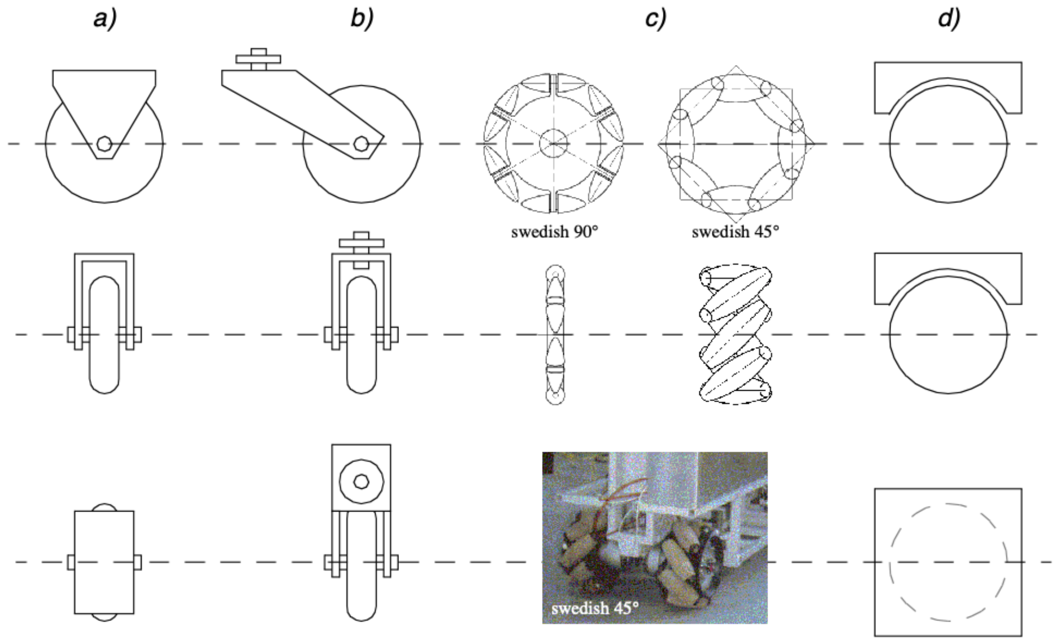


Figure 1.16: The four basic wheel types a)Standard wheel: Two degrees of freedom; rotation around the (motorized) wheel axle and the contact point b)castor wheel: Two degrees of freedom; rotation around an offset steering joint c)Swedish wheel: Three degrees of freedom; rotation around the (motorized) wheel axle, around the rollers and around the contact point

with the ground, while the castor wheel rotates around an offset axis, causing a force to be imparted to the robot chassis during steering.

The mecanum wheel[10] and the spherical wheel are both designs that are less constrained by directionality than the conventional standard wheel. The swedish wheel functions as a normal wheel, but provides low resistance in another direction as well, sometimes perpendicular to the conventional direction as in the swedish 90 and sometimes at an intermediate angle as in the swedish 45. The small rollers attached around the circumference of the wheel are passive and the wheel's primary axis serves as the only actively powered joint. The key advantage of this design is that, although the wheel rotation is powered only along the one principal axis (through the axle), the wheel can kinematically move with very little friction along many possible trajectories, not just forward and backward.

[10] Sometimes known as Swedish wheel or Ilon wheel after its inventor Bengt Erland Ilon [29].

The spherical wheel is a truly omnidirectional wheel, often designed so that it may be actively powered to spin along any direction. One mechanism for implementing this spherical design imitates the computer mouse, providing actively powered rollers that rest against the top surface of the sphere and impart rotational force.

Regardless of what wheel is used, in robots designed for all-terrain environments and in robots with more than three (3) wheels, a suspension system is normally required to maintain wheel contact with the ground. One of the simplest approaches to suspension is to design flexibility into the wheel itself. For instance, in the case of some four-wheeled indoor robots that use castor wheels, manufacturers have applied a deformeable tire of soft rubber to the wheel in order to create a primitive suspension. Of course, this limited solution cannot compete with a sophisticated suspension system in applications where the robot needs a more dynamic suspension for significantly non-flat terrain.

Wheel Geometry

The choice of wheel types for a mobile robot is strongly linked to the choice of wheel arrangement, or wheel geometry. When designing a mobile robot locomotion we must consider these two (2) issues simultaneously. Why does wheel type and wheel geometry matter? Three fundamental characteristics of a robot are governed by these choices:

- maneuverability,
- controllability
- stability.

Unlike automobiles, which are largely designed for a highly standardised environment[11] , mobile robots are designed for applications in a wide variety of situations. Automobiles all share similar wheel configurations as there is one region in the design space that maximises maneuverability, controllability and stability for their standard environment:

[11] such as the road network

the paved road.

However, there is no single wheel configuration that maximises these qualities for the variety of environments faced by different mobile robots. So, we will see great variety in the wheel configurations of mobile robots. In fact, few robots use the Ackerman wheel configuration of the automobile because of its poor maneuverability, with the exception of mobile robots designed for the road system (figure 2.20).

Table 2.1 gives an overview of wheel configurations ordered by the number of wheels. This table shows both the selection of particular wheel types and their geometric configuration on the robot chassis. Note that some of the configurations shown are of little use in mobile robot applications. For instance, the 2-wheeled bicycle arrangement has moderate maneu- verability and poor controllability. Like a single-leg hopping machine, it can never stand still. Nevertheless, this table provides an indication of the large variety of wheel configura- tions that are possible in mobile robot design.

1.3.2 Stability

Surprisingly, the minimum number of wheels required for static stability is two (2). As shown above, a two-wheel differential drive robot can achieve static stability if the center of mass is below the wheel axle. Cye is a commercial mobile robot that uses this wheel configuration

Figure 1.17: NAVLAB I, the first autonomous highway vehicle that steers and controls the throttle using vision and radar sensors [30].

However, under ordinary circumstances such a solution requires wheel diameters that are impractically large. Dynamics can also cause a two-wheeled robot to strike the floor with a third point of contact, for instance with sufficiently high motor torques from standstill. Conventionally, static stability requires a minimum of three (3)wheels, with the additional caveat that the center of gravity must be contained within the triangle formed by the ground contact points of the wheels. Stability can be further improved by adding more wheels, although once the number of contact points exceeds three, the hyperstatic nature of the geometry will require some form of flexible suspension on uneven terrain.

1.3.3 Manoeuvrability

Some robots are omnidirectional, meaning that they can move at any time in any direction along the ground plane (X, Y) regardless of the orientation of the robot around its vertical axis. This level of maneuverability requires wheels that can move in more than just a single direction, and so omnidirectional robots usually employ swedish or spherical wheels that are powered. A good example is Uranus, shown in figure 2.24. This robot uses four swedish wheels to rotate and translate independently and without constraints. In general, the ground clearance of robots with swedish and spherical wheels is somewhat limited, due to the mechanical constraints of constructing omnidirectional wheels. An interesting recent solution to the problem of omnidirectional navigation while solving this ground clearance problem is the four castor-wheeled configuration in which each castor wheel is actively steered and actively translated. In this configuration, the robot is truly omnidirectional because, even if the castor wheels are facing a direction perpendicular to the desired direction of travel, the robot can still move in the desired direction by steering these wheels. Because the vertical axis is offset from the ground contact path, the result of this steering motion is robot motion.

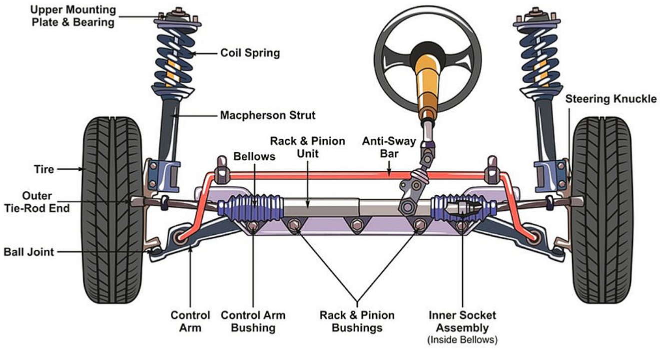


Figure 1.18: Example of an Ackerman drive used mostly in automotive industry [31].

In the research community, another classes of mobile robots are popular which achieve high maneuverability, only slightly inferior to that of the omnidirectional configurations. In such robots, motion in a particular direction may initially require a rotational motion. With a cir- cular chassis and an axis of rotation at the center of the robot, such a robot can spin without changing its ground footprint. The most popular such robot is the two-wheel differential drive robot where the two wheels rotate around the center point of the robot. One or two ad- ditional ground contact points may be used for stability, based on the application specifics.

In contrast to the above configurations, consider the Ackerman steering configuration com- mon in automobiles. Such a vehicle typically has a turning diameter that is larger than the car. Furthermore, for such a vehicle to move sideways requires a parking maneuver consist- ing of repeated changes in direction forward and backward. Nevertheless, Ackerman steer- ing geometries have been especially popular in the hobby robotics market, where a robot can be built by starting with a remote-control race car kit and adding sensing and autonomy to the existing mechanism. In addition, the limited maneuverability of Ackerman steering has an important advantage: its directionality and steering geometry provide it with very good lateral stability in high speed turns.

1.3.4 Controllability

There is generally an inverse correlation between controllability and maneuverability. For example, the omni-directional designs such as the four castor-wheeled configuration require significant processing to convert desired rotational and translational velocities to individual wheel commands. Furthermore, such omni-directional designs often have greater degrees of freedom at the wheel. For instance, the swedish wheel has a set of free rollers along the wheel perimeter. These degrees of freedom cause an accumulation of slippage, tend to reduce dead-reckoning accuracy and increase the design complexity.

Controlling an omnidirectional robot for a specific direction of travel is also more difficult and often less accurate when compared to less manoeuvrable designs.

> For example, an Ackerman steering vehicle can go straight simply by locking the steerable wheels and driving the drive wheels, which can be seen in **Fig.** 1.18.

In a differential drive vehicle, the two (2) motors attached to the two wheels must be driven along exactly the same velocity profile, which can be challenging considering variations between wheels, motors and environmental differences. With four-wheel om- nidrive, such as the Uranus robot which has four swedish wheels, the problem is even harder because all four wheels must be driven at exactly the same speed for the robot to travel in a perfectly straight line.

In summary, there is **NO** "ideal" drive configuration that simultaneously maximises stability, manoeuvrability and controllability. Each mobile robot application places unique constraints on the robot design problem, and the designer's task is to choose the most appropriate drive configuration possible from among this space of compromises.

1.3.5 Case Studies for Wheeled Motion

Now let's describe four (4) specific wheel configurations, in order to demonstrate concrete applications of the concepts above to mobile robots built for real-world activities.

Synchro Drive

The synchro drive configuration (figure 2.22) is a popular arrangement of wheels in indoor mobile robot applications. It is an interesting configuration because, although there are three driven and steered wheels, only two motors are used in total. The one translation motor sets the speed of all three wheels together, and the one steering motor spins all the wheels together about each of their individual vertical steering axes. But note that the wheels are being steered with respect to the robot chassis, and therefore there is no direct way of re-orienting the robot chassis. In fact, the chassis orientation does drift over time due to uneven tire slippage, causing rotational dead-reckoning error.

Synchro drive is particularly advantageous in cases where omnidirectionality is needed. So long as each vertical steering axis is aligned with the contact path of each tire, the robot can always re-orient its wheels and move along a new trajectory without changing its footprint. Of course, if the robot chassis has directionality and the designers intend to re-orient the chassis purposefully, then synchro drive is only appropriate when combined with an independently rotating turret that attaches to the wheel chassis. Commercial research robots such as the Nomadics 150 or the RWI B21r have been sold with this configuration (figure 1.12). In terms of dead-reckoning, synchro drive systems are generally superior to true omni-directional configurations but inferior to differential drive and Ackerman steering systems. There are two main reasons for this. First and foremost, the translation motor generally drives the three wheels using a single belt. Due to slop and backlash in the drivetrain, whenever the drive motor engages, the closest wheel begins spinning before the furthest wheel, causing a small change in the orientation of the chassis. With additional changes in motor speed, these small angular shifts accumulate to create a large error in orientation during dead-reckoning.

Second, the mobile robot has no direct control over the orientation of the chassis. Depending on the orientation of the chassis, the wheel thrust can be highly asymmetric, with two wheels on one side and the third wheel alone, or symmetric, with one wheel on each side and one wheel straight ahead or behind, as shown in (2.22). The asymmetric cases results in a variety of errors when tire-ground slippage can occur, again causing errors in dead-reckoning of robot orientation.

Omnidirectional Drive

As we will see later, omni-directional movement is of great interest for complete manoeuvrability. Omnidirectional robots which are able to move in any direction (x, y, θ) at any time are also holonomic. They can be realized by either using spheric, castor or swedish wheels.

Let's look at these three examples:

Omnidirectional locomotion with three spheric wheels The omnidirectional robot depicted in figure 2.23 is based on three spheric wheels, each actuated by one motor. In this design, the spheric wheels are suspended by three contact points, two given by spherical bearings and one by the a wheel connected to the motor axle. This concept provides excellent maneuverability and is simple in design. However, it is limited to flat surfaces and small loads, and it is quite difficult to find round wheels with high friction coefficients

Omnidirectional locomotion with four swedish wheels The omnidirectional arrangement depicted in figure 2.24 has been used successfully on several research robots, including the CMU Uranus. This configuration consists of four swedish 45 degree wheels, each driven by a separate motor. By varying the direction of rotation and relative speeds of the four wheels, the robot can be moved along any trajectory in the plane and, even more impressively, can simultaneously spin around its vertical axis. For example, when all four wheels spin "forward" or "backward", the robot as a whole moves in a straight line forward and backward, respectively. However, when one diagonal pair of wheels is spun in the same direction and the other diagonal pair is spun in the opposite direction, the robot moves laterally. This four-wheel arrangement of swedish wheels is not minimal in terms of control motors. Because there are only 3 degrees of freedom in the plane, one can build a three-wheeled om- nidirectional robot chassis using three swedish 90 degree wheels as shown in Table 2.1. However, existing examples such as Uranus have been designed with four wheels due to ca- pacity and stability considerations. One application for which such omnidirectional designs are particular amenable is mobile manipulation. In this case, it is desirable to reduce the degrees of freedom of the manipulator arm to save arm mass by using the mobile robot chassis motion for gross motion. As with humans, it would be ideal if the base could move omnidirectionally without greatly impact-

Omnidirectional locomotion with four castor wheels and eight motors Another solution for omnidirectionality is to use castor wheels. This is done for the Nomad XR4000 form Nomadics (fig. 2.25) giving it an excellent maneuverability. Unfortunately Nomadics Technology has ceased the production of mobile robots. The above two examples are drawn from Table 2.1, but this is not an exhaustive list of all wheeled locomotion techniques. Hybrid approaches that combine legged and wheeled loco- motion, or tracked and wheeled locomotion, can also offer particular advantages. Below are two unique designs created for specialized applications.

Tracked Slip/Skid Locomotion

In the wheel configurations discussed above, we have made the assumption that wheels are not allowed to skid against the surface. An alternative form of steering, termed slip/skid, may be used to re-orient the robot by spinning wheels that are facing the same direction at different speeds or in opposite directions. The army tank operates this way, and Nanokhod, pictured below (figure 2.26) is an example of a mobile robot based on the same concept. Robots that make use of tread have much larger ground contact patches, and this can signif- icantly improve their maneuverability in loose terrain compared to conventional wheeled designs. However, due to this large ground contact patch, changing the orientation of the ro- bot usually requires a skidding turn, wherein a large portion of the track must slide against the terrain. The disadvantage of such configurations is coupled to the slip/skid steering. Because of the large amount of skidding during a turn, the exact center of rotation of the robot is hard to predict and the exact change in position and orientation is also subject to variations depend- ing on the ground friction. Therefore, dead-reckoning on such robots is highly inaccurate. This is the trade-off that is made in return for extremely good maneuverability and traction over rough and loose terrain. Furthermore, a slip/skid approach on a high-friction surface can quickly overcome the torque capabilities of the motors being used. In terms of power efficiency, this approach is reasonably efficient on loose terrain but extremely inefficient otherwise.

1.3.6 Walking Wheels

Walking robots might offer the best maneuverability in rough terrain. However, they are in- efficient on flat ground and need sophisticated control. Hybrid solutions, combining the adaptability of legs with the efficiency of wheels offer an interesting compromise. Solutions that passively adapt to the terrain are of particular interest for field and space robotics. The Sojourner robot of NASA/JPL (fig. 1.2) represents such a hybrid solution, able to overcome objects up to the size of the wheels. A more advanced mobile robot design for similar appli- cations has recently been produced by EPFL (fig. 2.27). This robot, called Shrimp2, has 6 motorized wheels and is capable of climbing objects up to two times its wheel diameter [84,85]. This enables it to climb regular stairs though the robot is even smaller than the So- journer. Using a rhombus configuration, the Shrimp has a steering wheel in the front and the rear, and two wheels arranged on a bogy on each side. The front wheel has a spring suspen- sion to guarantee optimal ground contact of all wheels at any time. The steering of the rover is realized by synchronizing the steering of the front and rear wheels and the speed differ- ence of the bogy wheels. This allows for high precision maneuvers and turning on the spot with minimum slip/skid of the four center wheels. The use of parallel articulations for the front wheel and the bogies creates a virtual center of rotation at the level of the wheel axis. This ensures maximum stability and climbing abilities even for very low friction coeffi- cients between the wheel and the ground. As mobile robotics research matures we find ourselves able to design more intricate me- chanical systems. At the same time, the control problems of inverse kinematics and dynam2 Locomotion 37 R. Siegwart, EPFL, Illah Nourbakhsh, CMU ics are now so readily conquered that these complex mechanics can in general be controlled. So, in the near future, you should expect to see a great number of unique, hybrid mobile ro- bots that draw together advantages from several of the underlying locomotion mechanisms that we have discussed in this chapter. They will each be technologically impressive, and each will be designed as the expert robot for its particular environmental niche.

Chapter 2

Perception

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2.1 Introduction

One of the most important tasks of an AMR is to acquire knowledge about its environment.[1] This is achieved by taking measurements using various sensors and then extracting meaningful information from those measurements.

[1]One could even argue it is the definition of life, if you ask a biologist as the ability to feel and act on its environment is the bare necessity.

In this chapter we present the most common sensors used in AMR and then discuss strategies for extracting information from the sensors.

2.1.1 Sensors for Mobile Robotics

There is a wide variety of sensors used in AMRs (Fig. 4.1). Some are used to measure simple values like the internal temperature of a robot's electronics or the rotational speed of the motors in its wheels or actuators. Other, more sophisticated sensors can be used to acquire information about the robot's environment or even to directly measure a robot's global position. Here, we focus primarily on sensors used to extract information about the robot's environment. Because a AMR moves around, it will frequently encounter unforeseen environmental characteristics, and therefore such sensing is particularly critical. We begin with a functional classification of sensors. Then, after presenting basic tools for describing a sensor's performance, we proceed to describe selected sensors in detail.

2.1.2 Sensor Classification

We classify sensors using two (2) important functional axes. Let's define these terms for clarity;

Proprioceptive sensors which measure values internal to the robot.

e.g., motor speed, wheel load, robot arm joint angles, battery voltage.

Exteroceptive sensors which measure information from the robot's environment;

e.g., distance measurements, light intensity, sound amplitude.

> exteroceptive sensor measurements are interpreted by the robot to extract meaningful environmental features.

Passive sensors measure ambient environmental energy entering the sensor.

e.g., temperature probes, microphones and CCD or CMOS cameras.

Active sensors emit energy into the environment, then measure the environmental reaction. Because active sensors can manage more controlled interactions with the environment, they often achieve superior performance. However, active sensing introduces several risks: the outbound energy may affect the very characteristics that the sensor is attempting to measure. Furthermore, an active sensor may suffer from interference between its signal and those beyond its control. For example, signals emitted by other nearby robots, or similar sensors on the same robot my influence the resulting measurements. Examples of active sensors include wheel quadrature encoders, ultrasonic sensors and laser rangefinders.

The sensor classes in Table (4.1) are arranged in ascending order of complexity and descending order of technological maturity. Tactile sensors and proprioceptive sensors are critical to virtually all mobile robots, and are well understood and easily implemented. Commercial quadrature encoders, for example, may be purchased as part of a gear-motor assembly used in a AMR. At the other extreme, visual interpretation by means of one or more CCD/CMOS cameras provides a broad array of potential functionalities, from obstacle avoidance and localisation to human face recognition. However, commercially available sensor units that provide visual functionalities are only now beginning to emerge

2.1.3 Characterising Sensor Performance

The sensors we describe in this chapter vary greatly in their performance characteristics. Some sensors provide extreme accuracy in well-controlled laboratory settings, but are overcome with error when subjected to real-world environmental variations. Other sensors provide narrow, high precision data in a wide variety settings. To quantify such performance characteristics, first we formally define the sensor performance terminology that will be valuable throughout the rest of this chapter.

Basic Sensor Response Ratings

A number of sensor characteristics can be rated quantitatively in a laboratory setting. Such performance ratings will necessarily be best-case scenarios when the sensor is placed on a real-world robot, but are nevertheless useful.

Dynamic Range Used to measure the spread between the lower and upper limits of inputs values to the sensor while maintaining normal sensor operation. Formally, the dynamic range is the ratio of the maximum input value to the minimum measurable input value. Because this raw ratio can be unwieldy, it is usually measured in Decibels, which is computed as ten times the common logarithm of the dynamic range. However, there is potential confusion in the calculation of Decibels, which are meant to measure the ratio between powers, such as Watts or Horsepower.

Suppose your sensor measures motor current and can register values from a minimum of 1 mA to 20 A. The dynamic range of this current sensor is defined as:

$$10 \cdot \log \left[\frac{20}{0.001} \right] = 43 \quad \text{dB} \tag{2.1}$$

Now suppose you have a voltage sensor that measures the voltage of your robot's battery, measuring any value from 1 mV to 20 V. Voltage is **NOT** a unit of power, but the square of voltage is proportional to power. Therefore, we use 20 instead of 10:

$$20 \cdot \log \left[\frac{20}{0.001} \right] = 86 \quad \text{dB} \tag{2.2}$$

Range An important rating in AMR because often robot sensors operate in environments where they are frequently exposed to input values beyond their working range. In such cases, it is critical to understand how the sensor will respond. For example, an optical rangefinder will have a minimum operating range and can thus provide spurious data when measurements are taken with object closer than that minimum.

Resolution The minimum difference between two (2) values that can be detected by a sensor. Usually, the lower limit of the dynamic range of a sensor is equal to its resolution. However, in the case of digital sensors, this is not necessarily so. For example, suppose that you have a sensor that measures voltage, performs an analogue-to-digital conversion and outputs the converted value as an 8-bit number linearly corresponding to between 0 and 5 Volts. If this sensor is truly linear, then it has $2^8 - 1$ total output values or a resolution of:

$$\frac{5}{255} = 20 \quad \text{mV}$$

Linearity is an important measure governing the behaviour of the sensor's output signal as the input signal varies. A linear response indicates that if two (2) inputs, say x and y result in the two outputs $f(x)$ and $f(x)$, then for any values a and b, the following relation can be derived:

$$f(x + y) = f(x) + f(y) .$$

This means that a plot of the sensor's input/output response is simply a straight line.

Bandwidth or Frequency is used to measure the speed with which a sensor can provide a stream of readings. Formally, the number of measurements per second is defined as the sensor's frequency in Hz. Because of the dynamics of moving through their environment, mobile robots often are limited in maximum speed by the bandwidth of their obstacle detection sensors. Thus increasing the bandwidth of ranging and vision-based sensors has been a high-priority goal in the robotics community.

In Situ Sensor Performance

The above sensor characteristics can be reasonably measured in a laboratory environment, with confident extrapolation to performance in real-world deployment. However, a number of important measures cannot be reliably acquired without deep understanding of the complex interaction between all environmental characteristics and the sensors in question. This is most relevant to the most sophisticated sensors, including active ranging sensors and visual interpretation sensors.

Sensitivity A measure of the degree to which an incremental change in the target input signal changes the output signal. Formally, sensitivity is the ratio of output change to input change. Unfortunately, however, the sensitivity of exteroceptive sensors is often confounded by undesirable sensitivity and performance coupling to other environmental parameters.

Cross-Sensitivity is the technical term for sensitivity to environmental parameters that are orthogonal to the target parameters for the sensor. For example, a flux-gate compass can demonstrate high sensitivity to magnetic north and is therefore of use for AMR navigation. However, the compass will also demonstrate high sensitivity to ferrous building materials, so much so that its cross-sensitivity often makes the sensor useless in some indoor environments. High cross-sensitivity of a sensor is generally undesirable, especially so when it cannot be modelled.

Error of a sensor is defined as the difference between the sensor's output measurements and the true values being measured, within some specific operating context.

As an example, given a true value v and a measured value m, we can define error as:

$$\text{Error} = m - v .$$

Accuracy defined as the degree of conformity between the sensor's measurement and the true value, and is often expressed as a proportion of the true value (e.g. 97.5% accuracy):

$$\text{Accuracy} = 1 - \frac{|m - v|}{v}.$$

Of course, obtaining the ground truth (v), can be difficult or impossible, and so establishing a confident characterisation of sensor accuracy can be problematic. Further, it is important to distinguish between two different sources of error:

- Systematic errors are caused by factors or processes that can in theory be modelled. These errors are, therefore, deterministic.[2]

  Poor calibration of a laser rangefinder, un-modelled slope of a hallway floor and a bent stereo camera head due to an earlier collision are all possible causes of systematic sensor errors

- Random errors cannot be predicted using a sophisticated model nor can they be mitigated with more precise sensor machinery. These errors can only be described in probabilistic terms (i.e. stochastic). Hue instability in a colour camera, spurious range-finding errors and black level noise in a camera are all examples of random errors.

[2]Meaning, it's value is not determined by a random process and therefore should, in theory, be predictable.

Precision is often confused with accuracy, and now we have the tools to clearly distinguish these two terms. Intuitively, high precision relates to reproducibility of the sensor results. For example, one sensor taking multiple readings of the same environmental state has high precision if it produces the same output. In another example, multiple copies of this sensors taking readings of the same environmental state have high precision if their outputs agree. Precision does not, however, have any bearing on the accuracy of the sensor's output with respect to the true value being measured. Suppose that the random error of a sensor is characterised by some mean value (μ) and a standard deviation (σ). The formal definition of precision is the ratio of the sensor's output range to the standard deviation:

$$\text{Precision} = \frac{\text{Range}}{\sigma}.$$

> Only σ and **NOT** μ has impact on precision. In contrast mean error is directly proportional to overall sensor error and inversely proportional to sensor accuracy.

Characterising Error

Mobile robots depend heavily on exteroceptive sensors. Many of these sensors concentrate on a central task for the robot:

acquiring information on objects in the robot's immediate vicinity so that it may interpret the state of its surroundings.

Of course, these "objects" surrounding the robot are all detected from the viewpoint of its local reference frame.[3] Since the systems we study are mobile, their ever-changing position and their motion has a significant impact on overall sensor behaviour.

[3] In this case we are referring to the robot reference frame.

Now that we have the necessary knowledge on the fundamental concepts and terminology, we can now describe how dramatically the sensor error of an AMR disagrees with the ideal picture drawn in the previous section.

Blurring of Systematical and Random Errors

Active ranging sensors tend to have failure modes which are triggered largely by specific relative positions of the sensor and environment targets.

> For example, a sonar sensor will product specular reflections,[4] producing grossly inaccurate measurements of range, at specific angles to a smooth sheet-rock wall.

[4] The incident light is reflected into a single outgoing direction.

During motion of the robot, such relative angles occur at stochastic intervals. This is especially true in a AMR outfitted with a ring of multiple sonars. The chances of one sonar entering this error mode during robot motion is high. From the perspective of the moving robot, the sonar measurement error is a random error in this case. However, if the robot were to stop, becoming motionless, then a very different error modality is possible.

If the robot's static position causes a particular sonar to fail in this manner, the sonar will fail consistently and will tend to return precisely the same (and incorrect!) reading time after time. Once the robot is motionless, the error appears to be systematic and high precision.

> The fundamental mechanism at work here is the cross-sensitivity of AMR sensors to robot pose and robot-environment dynamics.

The models for such cross-sensitivity are **NOT**, in an underlying sense, truly random. However, these physical interrelationships are rarely modelled and therefore, from the point of view of an incomplete model, the errors appear random during motion and systematic when the robot is at rest. Sonar is not the only sensor subject to this blurring of systematic and random error modality. Visual interpretation through the use of a CCD camera is also highly susceptible to robot motion and position because of camera dependency on lighting.[5]

[5] such as glare and reflections.

> The important point is to realise that, while systematic error and random error are well-defined in a controlled setting, the AMR can exhibit error characteristics that bridge the gap between deterministic and stochastic error mechanisms.

Multi-Modal Error Distributions

It is common to characterise the behaviour of a sensor's random error in terms of a probability distribution over various output values. In general, one knows very little about the causes of random error and therefore several simplifying assumptions are commonly used. For example, we can assume that the error is zero-mean ($\mu = 0$), in that it symmetrically generates both positive and negative measurement error. We can go even further and assume that the probability density curve is Gaussian. Although we discuss the mathematics of this in detail later, it is important for now to recognise the fact that one frequently assumes symmetry as well as unimodal distribution. This means that measuring the correct value is most probable, and any measurement that is further away from the correct value is less likely than any measurement that is closer to the correct value. These are strong assumptions that enable powerful mathematical principles to be applied to AMR problems, but it is important to realise how wrong these assumptions usually are.

Consider, for example, the sonar sensor once again. When ranging an object that reflects the sound signal well, the sonar will exhibit high accuracy, and will induce random error based on noise, for example, in the timing circuitry. This portion of its sensor behaviour will exhibit error characteristics that are fairly symmetric and unimodal. However, when the sonar sensor is moving through an environment and is sometimes faced with materials that cause coherent reflection rather than returning the sound signal to the sonar sensor, then the sonar will grossly overestimate distance to the object. In such cases, the error will be biased toward positive measurement error and will be far from the correct value. The error is not strictly systematic, and so we are left modelling it as a probability distribution of random error. So the sonar sensor has two (2) separate types of operational modes, one in which the signal does return and some random error is possible, and the second in which the signal returns after a multi-path reflection, and gross overestimation error occurs. The probability distribution could easily be at least bimodal in this case, and since overestimation is more common than underestimation it will also be asymmetric.

As a second example, consider ranging via stereo vision. Once again, we can identify two (2) modes of operation. If the stereo vision system correctly correlates two images, then the resulting random error will be caused by camera noise and will limit the measurement accuracy. But the stereo vision system can also correlate two images incorrectly, matching two fence posts for example that are not the same post in the real world. In such a case stereo vision will exhibit gross measurement error, and one can easily imagine such behaviour violating both the unimodal and the symmetric assumptions. The thesis of this section is that sensors in a AMR may be subject to multiple modes of operation and, when the sensor error is characterised, uni modality and symmetry may be grossly violated. Nonetheless, as you will see, many successful AMR systems make use of these simplifying assumptions and the resulting mathematical techniques with great empirical success. The above sections have presented a terminology with which we can characterise the advantages and disadvantages of various mobile robot sensors. In the following sections, we do the same for a sampling of the most commonly used AMR sensors today.

2.1.4 Wheel and Motor Sensors

Wheel/motor sensors are devices use to measure the internal state and dynamics of a mobile robot. These sensors have vast applications outside of AMR and, as a result, AMR has enjoyed the benefits of high-quality, low-cost wheel and motor sensors which offer excellent resolution.

In the next part, we sample just one such sensor, the optical incremental encoder.

Optical Encoders

Optical incremental encoders have become the most popular device for measuring angular speed and position within a motor drive or at the shaft of a wheel or steering mechanism. In mobile robotics, encoders are used to control the position or speed of wheels and other mo- tor-driven joints. Because these sensors are proprioceptive, their estimate of position is best in the reference frame of the robot and, when applied to the problem of robot localisation, significant corrections are required as discussed in Chapter 5.

An optical encoder is basically a mechanical light chopper that produces a certain number of sine or square wave pulses for each shaft revolution. It consists of an illumination source, a fixed grating that masks the light, a rotor disc with a fine optical grid that rotates with the shaft, and fixed optical detectors. As the rotor moves, the amount of light striking the optical detectors varies based on the alignment of the fixed and moving gratings. In robotics, the resulting sine wave is transformed into a discrete square wave using a threshold to choose between light and dark states. Resolution is measured in Cycles Per Revolution (CPR).

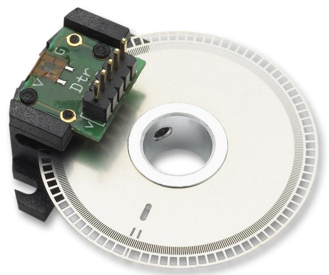


Figure 2.1: An example of a rotary encoder. [32]

The minimum angular resolution can be readily computed from an encoder's CPR rating. A typical encoder in AMR may have 2,000 CPR while the optical encoder industry can readily manufacture encoders with 10,000 CPR. In terms of required bandwidth, it is of course critical that the encoder be sufficiently fast to count at the shaft spin speeds that are expected. Industrial optical encoders present no bandwidth limitation to AMR applications. Usually in AMR the quadrature encoder is used. In this case, a second illumination and detector pair is placed 90ř shifted with respect to the original in terms of the rotor disc. The resulting twin square waves, shown in Fig. 4.2, provide significantly more information. The ordering of which square wave produces a rising edge first identifies the direction of rotation. Furthermore, the four detectability different states improve the resolution by a factor of four with no change to the rotor disc. Thus, a 2,000 CPR encoder in quadrature yields 8,000 counts. Further improvement is possible by retaining the sinusoidal wave measured by the optical detectors and performing sophisticated interpolation. Such methods, although rare in AMR, can yield 1000-fold improvements in resolution. As with most proprioceptive sensors, encoders are generally in the controlled environment of a AMR's internal structure, and so systematic error and cross-sensitivity can be engineered away. The accuracy of optical encoders is often assumed to be

100% and, although this may not entirely correct, any errors at the level of an optical encoder are dwarfed by errors downstream of the motor shaft.

Heading Sensors

Heading sensors can be proprioceptive (gyroscope, inclinometer) or exteroceptive (com- pass). They are used to determine the robots orientation and inclination. They allow us, to- gether with appropriate velocity information, to integrate the movement to a position estimate. This procedure, which has its roots in vessel and ship navigation, is called dead reckoning.

Compasses

The two most common modern sensors for measuring the direction of a magnetic field are the Hall Effect and Flux Gate compasses. Each has advantages and disadvantages, as described below. The Hall Effect describes the behaviour of electric potential in a semiconductor when in the presence of a magnetic field. When a constant current is applied across the length of a semi- conductor, there will be a voltage difference in the perpendicular direction, across the semi- conductor's width, based on the relative orientation of the semiconductor to magnetic flux lines. In addition, the sign of the voltage potential identifies the direction of the magnetic field. Thus, a single semiconductor provides a measurement of flux and direction along one dimension. Hall Effect digital compasses are popular in AMR, and contain two such semiconductors at right angles, providing two axes of magnetic field (thresholded) direction, thereby yielding one of 8 possible compass directions. The instruments are inexpensive but also suffer from a range of disadvantages. Resolution of a digital hall effect compass is poor. Internal sources of error include the nonlinearity of the basic sensor and systematic bias errors at the semiconductor level. The resulting circuitry must perform significant filtering, and this lowers the bandwidth of hall effect compasses to values that are slow in AMR terms. For example the hall effect compasses pictured in figure 4.3 needs 2.5 seconds to settle after a 90ř spin. The Flux Gate compass operates on a different principle. Two small coils are wound on fer- rite cores and are fixed perpendicular to one-another. When alternating current is activated in both coils, the magnetic field causes shifts in the phase depending upon its relative alignment with each coil. By measuring both phase shifts, the direction of the magnetic field in two dimensions can be computed. The flux-gate compass can accurately measure the strength of a magnetic field and has improved resolution and accuracy; however it is both larger and more expensive than a Hall Effect compass. Regardless of the type of compass used, a major drawback concerning the use of the Earth's magnetic field for AMR applications involves disturbance of that magnetic field by other magnetic objects and man-made structures, as well as the bandwidth limitations of electronic compasses and their susceptibility

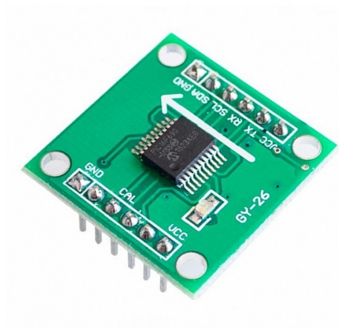


Figure 2.2: An example of an electronic compass [33].

to vibration. Particularly in indoor environments AMR applications have often avoided the use of compasses, although a compass can conceivably provide useful local orientation information indoors, even in the precense of steel structures.

Gyroscope

Gyroscopes are heading sensors which preserve their orientation in relation to a fixed refer- ence frame. Thus they provide an absolute measure for the heading of a mobile system. Gy- roscopes can be classified in two categories, mechanical gyroscopes and optical gyroscopes.

Mechanical Gyroscopes

The concept of a mechanical gyroscope relies on the inertial properties of a fast spinning rotor. The property of interest is known as the gyroscopic precession. If you try to rotate a fast spinning wheel around its vertical axis, you will feel a harsh reaction in the horizontal axis. This is due to the angular momentum associated with a spinning wheel and will keep the axis of the gyroscope inertially stable. The reactive torque τ and thus the tracking stability with the inertial frame are proportional to the spinning speed ω, the precession speed Ω and the wheel's inertia I.

$$\tau = I\omega\Omega$$

By arranging a spinning wheel as seen in Figure 4.4, no torque can be transmitted from the outer pivot to the wheel axis. The spinning axis will therefore be space-stable (i.e. fixed in an inertial reference frame). Nevertheless, the remaining friction in the bearings of the gyro- axis introduce small torques, thus limiting the long term space stability and introducing small errors over time. A high quality mechanical gyroscope can cost up to $100,000 and has an angular drift of about 0.1ř in 6 hours. For navigation, the spinning axis has to be initially selected. If the spinning axis is aligned with the north-south meridian, the earth's rotation has no effect on the gyro's horizontal axis. If it points east-west, the horizontal axis reads the earth rotation. Rate gyros have the same basic arrangement as shown in Figure 4.4 but with a slight modification. The gimbals are restrained by a torsional spring with additional viscous damping. This enables the sensor to measure angular speeds instead of absolute orientation.

Optical Gyroscopes

Optical gyroscopes are a relatively new innovation. Commercial use began in the early 1980's when they were first installed in aircraft. Optical gyroscopes are angular speed sen- sors that use two monochromatic light beams, or lasers, emitted from the same source in- stead of moving, mechanical parts. They work on the principle that the speed of light remains unchanged and, therefore, geometric change can cause light to take a varying amount of time to reach its destination. One laser beam is sent traveling clockwise through a fiber while the other travels counterclockwise. Because the laser

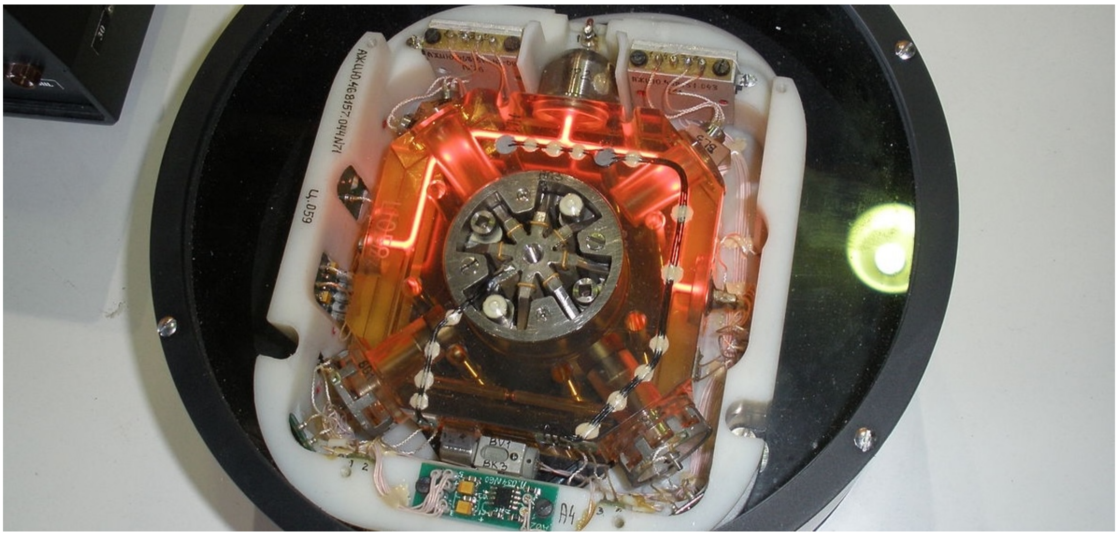


Figure 2.3: Optical Gyroscopes have no moving parts, (unlike mechanical gyroscopes) making them extremely reliable [34].

traveling in the direction of rotation has a slightly shorter path, it will have a higher frequency. The difference in fre- quency of the two beams is a proportional to the angular velocity of the cylinder. New solid-state optical gyroscopes based on the same principle are build using microfabrication technology, thereby providing heading information with resolution and bandwidth far beyond the needs of mobile robotic applications. Bandwidth, for instance, can easily exceed 100KHz while resolution can be smaller than 0.0001ř/hr.

Ground Based Beacons

Figure 2.4

One elegant approach to solving the localization problem in AMR is to use active or passive beacons. Using the interaction of on-board sensors and the environmental bea- cons, the robot can identify its position precisely. Although the general intuition is identical to that of early human navigation beacons, such as stars, mountains and lighthouses, modern technology has enabled sensors to localize an outdoor robot with accuracies of better than 5 cm within areas that are kilometres in size.

In the following subsection, we describe one such beacon system, the Global Positioning System (GPS), which is extremely effective for outdoor ground-based and flying robots. In- door beacon systems have been generally less successful for a number of reasons. The ex- pense of environmental modification in an indoor setting is not amortized over an extremely large useful area, as it is for example in the case of GPS. Furthermore, indoor environments offer significant challenges not seen outdoors, including multipath and environment dynam- ics. A laser-based indoor beacon system, for example, must disambiguate the one true laser signal from possibly tens of other powerful signals that have reflected off of walls, smooth floors and doors. Confounding this, humans and other obstacles may be constantly changing the environment, for example occluding the one true path from the beacon to the robot. In commercial applications such as manufacturing plants, the environment can be carefully controlled to ensure success. In less structured indoor settings, beacons have nonetheless been used, and the problems are mitigated by careful beacon placement and the useful of passive sensing modalities.

Global Positioning System

The Global Positioning System (GPS) was initially developed for military use but is now freely available for civilian navigation. There are at least 24 operational GPS satellites at all times. The satellites orbit every 12 hours at a height of 20.190km. There are four (4) satellites which located in each of six planes inclined 55° with respect to the plane of the earth's equator (figure 4.5).

Each satellite continuously transmits data which indicates its location and the current time. Therefore, GPS receivers are completely passive but exteroceptive sensors. The GPS satellites synchronise their transmissions to allow their signals to be sent at the same time. When a GPS receiver reads the transmission of two (2) or more satellites, the arrival time differences inform the receiver as to its relative distance to each satellite.

By combining information regarding the arrival time and instantaneous location of four (4) satellites, the receiver can infer its own position.

> In theory, such triangulation requires only three (3) data points. However, timing is extremely critical in the GPS application because the time intervals being measured are in ns.

It is, of course, mandatory the satellites to be well synchronised. To this end, they are updated by ground stations regularly and each satellite carries on-board atomic clocks[6] for timing. The GPS receiver clock is also important so that the travel time of each satellite's transmission can be accurately measured. But GPS receivers have a simple quartz clock. So, although 3 satellites would ideally provide position in three axes, the GPS receiver requires 4 satellites, using the additional information to solve for 4 variables: three position axes plus a time correction. The fact that the GPS receiver must read the transmission of 4 satellites simultaneously is a significant limitation. GPS satellite transmissions are extremely low-power, and reading them successfully requires direct

[6]An example of a csium clock for use in GPS.

line-of-sight communication with the satellite. Thus, in confined spaces such as city blocks with tall buildings or dense forests, one is unlikely to receive 4 satellites reliably. Of course, most indoor spaces will also fail to provide sufficient visibility of the sky for a GPS receiver to function. For these reasons, GPS has been a pop- ular sensor in AMR, but has been relegated to projects involving AMR traversal of wide-open spaces and autonomous flying machines. A number of factors affect the performance of a localization sensor that makes use of GPS. First, it is important to understand that, because of the specific orbital paths of the GPS sat- ellites, coverage is not geometrically identical in different portions of the Earth and therefore resolution is not uniform. Specifically, at the North and South poles, the satellites are very close to the horizon and, thus, while resolution in the latitude and longitude directions is good, resolution of altitude is relatively poor as compared to more equatorial locations.

The second point is that GPS satellites are merely an information source. They can be employed with various strategies in order to achieve dramatically different levels of localisation resolution. The basic strategy for GPS use, called pseudorange and described above, generally performs at a resolution of 15m. An extension of this method is differential GPS, which makes use of a second receiver that is static and at a known exact position. A number of errors can be corrected using this reference, and so resolution improves to the order of 1m or less. A disadvantage of this technique is that the stationary receiver must be installed, its location must be measured very carefully and of course the moving robot must be within ki- lometers of this static unit in order to benefit from the DGPS technique. A further improved strategy is to take into account the phase of the carrier signals of each received satellite transmission. There are two carriers, at 19cm and 24cm, therefore significant improvements in precision are possible when the phase difference between multiple satellites is measured successfully. Such receivers can achieve 1cm resolution for point po- sitions and, with the use of multiple receivers as in DGPS, sub-1cm resolution. A final consideration for AMR applications is bandwidth. GPS will generally offer no better than 200 - 300ms latency, and so one can expect no better than 5Hz GPS updates. On a fast-moving AMR or flying robot, this can mean that local motion integration will be required for proper control due to GPS latency limitations.

2.2 Active Ranging

Active range sensors continue to be the most popular sensors used in AMR. Many ranging sensors have a low price point, and most importantly all ranging sensors provide easily interpreted outputs:

> Direct measurements of distance from the robot to objects in its vicinity.

For obstacle detection and avoidance, most AMR rely heavily on active ranging sensors. But the local free-space information provided by range sensors can also be accumulated into representations beyond the robot's current local reference frame. Therefore, active range sensors are also commonly found as part of the localisation and environmental modelling processes of AMRs.

It is only with the slow advent of successful visual interpretation competency that we can expect the class of active ranging sensors to gradually lose their primacy as the sensor class of choice among AMR engineers.

Below, we present two (2) Time-of-Flight (ToF) active range sensors:

- the ultrasonic sensor,
- the laser rangefinder.

Continuing onwards, we then present two (2) geometric active range sensors:

- the optical triangulation sensor,
- the structured light sensor.

Time-of-FLight Active Ranging

ToF ranging makes use of the propagation speed of sound or an electromagnetic wave. In general, the travel distance of a sound of electromagnetic wave is given by:

$$d = ct,$$

where d is the distance travelled usually round-trip (m), c the speed of wave propagation ($\text{m} \cdot \text{s}^{-1}$), and t is the time it takes to travel (s).

It is important to point out the propagation speed v of sound is approximately $0.3\ \text{m} \cdot \text{ms}^{-1}$ whereas the speed of an electromagnetic signal is $0.3\ \text{m} \cdot \text{ns}^{-1}$, which is one million times faster. The ToF for a typical distance, say 3 m, is 10 ms for an ultrasonic system but only 10 ns for a laser rangefinder. It is therefore obvious that measuring the time of flight t with electromagnetic signals is more technologically challenging.[7]

[7] This explains why laser range sensors have only recently become affordable and robust for use on mobile robots.

The quality of ToF range sensors depends mainly on the following:

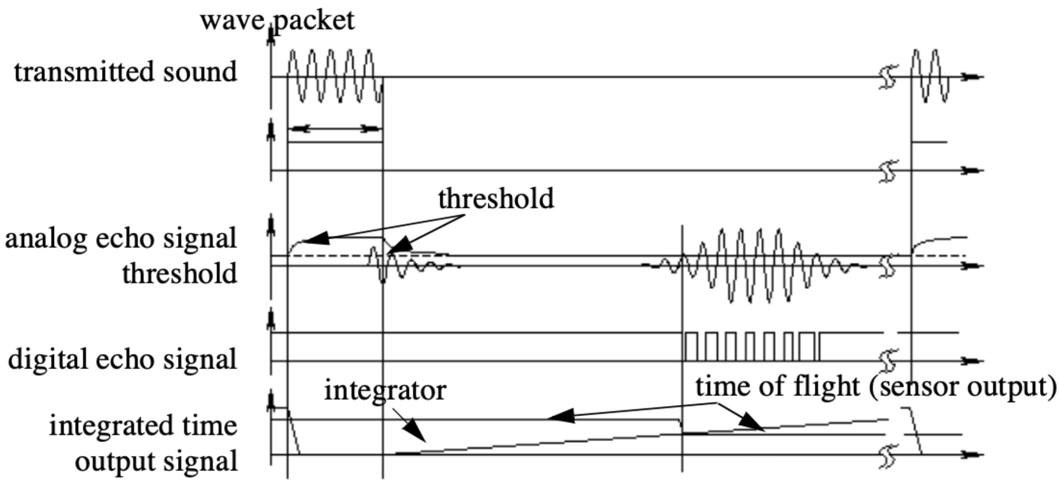


Figure 2.5: Signals of an ultrasonic sensor.

- Uncertainties in determining the exact time of arrival of the reflected signal,
- Inaccuracies in the time of flight measurement, particularly with laser range sensors,
- The dispersal cone of the transmitted beam mainly with ultrasonic range sensors
- Interaction with the target (e.g., surface absorption, specular reflections)
- Variation of propagation speed, and
- The speed of the AMR and target (in the case of a dynamic target).

As discussed below, each type of ToF sensor is sensitive to a particular subset of the above list of factors.

2.2.1 The Ultrasonic Sensor

The main ethos of an ultrasonic[8] sensor is to transmit a packet of ultrasonic pressure waves and to measure the time it takes for this wave to reflect and return to the receiver. The distance d of the object causing the reflection can be calculated based on the propagation speed of sound[9] c and the time of flight t.

$$d = \frac{c \times t}{2}$$

The speed of sound (v) in air is given by the following relation:

$$v = \sqrt{\gamma R T}$$

where γ is the ratio of specific heat, R is the gas constant $(\mathrm{J} \cdot \mathrm{mol}^{-1} \cdot \mathrm{K}^{-1})$, and T is the temperature

[8] Ultrasound is sound with frequencies greater than 20 kHz.

[9] Of course in this regard careful consideration needs to be made if the medium is significantly different than that of air (i.e., water).

in Kelvin (K). In air, at standard pressure, and 20 °C the speed of sound is approximately:

$$v = 343 \quad \mathrm{m \cdot s^{-1}}.$$

We can see the different signal output and input of an ultrasonic sensor in **Fig.** 2.5.

First, a series of sound pulses are emitted, which creates the wave packet. An integrator also begins to linearly climb in value, measuring the time from the transmission of these sound waves to detection of an echo. A threshold value is set for triggering an incoming sound wave as a valid echo.

> This threshold is often decreasing in time, because the amplitude of the expected echo decreases over time based on dispersal as it travels longer.

But during transmission of the initial sound pulses and just afterwards, the threshold is set very high to suppress triggering the echo detector with the outgoing sound pulses. A transducer will continue to ring for up to several ms after the initial transmission, and this governs the blanking time of the sensor.

> If, during the blanking time, the transmitted sound were to reflect off of an extremely close object and return to the ultrasonic sensor, it may fail to be detected.

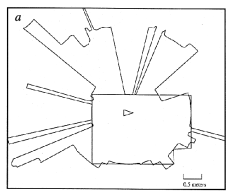
Figure 2.6: An example of an ultrasonic sensor used in Raspberry Pi applications [35].

However, once the blanking interval has passed, the system will detect any above-threshold reflected sound, triggering a digital signal and producing the distance measurement using the integrator value.

The ultrasonic wave typically has a frequency between 40 and 180 kHz and is usually generated by a piezo or electrostatic transducer. Often the same unit is used to measure the reflected signal, although the required blanking interval can be reduced through the use of separate output and input devices. Frequency can be used to select a useful range when choosing the appropriate ultrasonic sensor for a AMR. Lower frequencies correspond to a longer range, but with the disadvantage of longer post-transmission ringing and, therefore, the need for longer blanking intervals.

Most ultrasonic sensors used by AMRs have an effective range of roughly 12 cm to 5 metres. The published accuracy of commercial ultrasonic sensors varies between 98% and 99.1%. In AMR applications, specific implementations generally achieve a resolution of approximately 2 cm.

In most cases one may want a narrow opening angle for the sound beam in order to also obtain precise directional information about objects that are encountered. This is a major limitation since sound propagates in a cone-like manner with opening angles around 20° and 40°. Consequently, when using ultrasonic ranging one does not acquire depth data points but, rather, entire regions of constant depth. This means that the sensor tells us only that there is an object at a certain distance in within the area of the measurement cone. The sensor readings must be plotted as segments of an arc (sphere for 3D) and not as point measurements.[10] However, recent research developments



[10] The results of a 360° scan of a room.

show significant improvement of the measurement quality in using sophisticated echo processing. Ultrasonic sensors suffer from several additional drawbacks, namely in the areas of error, bandwidth and cross-sensitivity. The published accuracy values for ultrasonic sensors are nominal values based on successful, perpendicular reflections of the sound wave off an acoustically reflective material.

This does not capture the effective error modality seen on a AMR moving through its environment. As the ultrasonic transducer's angle to the object being ranged varies away from perpendicular, the chances become good that the sound waves will coherently reflect away from the sensor, just as light at a shallow angle reflects off of a mirror. Therefore, the true error behavior of ultrasonic sensors is compound, with a well-understood error distribution near the true value in the case of a successful retro-reflection, and a more poorly-understood set of range values that are grossly larger than the true value in the case of coherent reflection.

Of course the acoustic properties of the material being ranged have direct impact on the sensor's performance. Again, the impact is discrete, with one material possibly failing to produce a reflection that is sufficiently strong to be sensed by the unit. For example, foam, fur and cloth can, in various circumstances, acoustically absorb the sound waves. A final limitation for ultrasonic ranging relates to bandwidth. Particularly in moderately open spaces, a single ultrasonic sensor has a relatively slow cycle time.

For example, measuring the distance to an object that is 3 m away will take such a sensor 20ms, limiting its operating speed to 50 Hz. But if the robot has a ring of 20 ultrasonic sensors, each firing sequentially and measuring to minimize interference between the sensors, then the ring's cycle time becomes 0.4s and the overall update frequency of any one sensor is just 2.5 Hz. For a robot conducting moderate speed motion while avoiding obstacles using ultrasonic sensor, this update rate can have a measurable impact on the maximum speed possible while still sensing and avoiding obstacles safely.

> Ultrasonic measurements may be limited through barrier layers with large salinity, temperature or vortex differentials.

Laser Rangefinder

The laser rangefinder is a ToF sensor which achieves significant improvements over the ultrasonic range sensor due to the use of laser light instead of sound. This type of sensor consists of a transmitter which illuminates a target with a collimated[11] beam (e.g. laser), and a receiver capable of detecting the component of light which is essentially coaxial with the transmitted beam. Often referred to as optical radar or Light Detection and Ranging (LIDAR), these devices produce a range estimate based on the time needed for the light to reach the target and return.

[11] meaning all the rays in questions are made accurately parallel.

A mechanical mechanism with a mirror sweeps the light beam to cover the required scene in a plane or even in 3 dimensions, using a rotating mirror. One way to measure the ToF for the light beam is to use a pulsed laser and then measured the elapsed time directly, just as in the ultrasonic solution

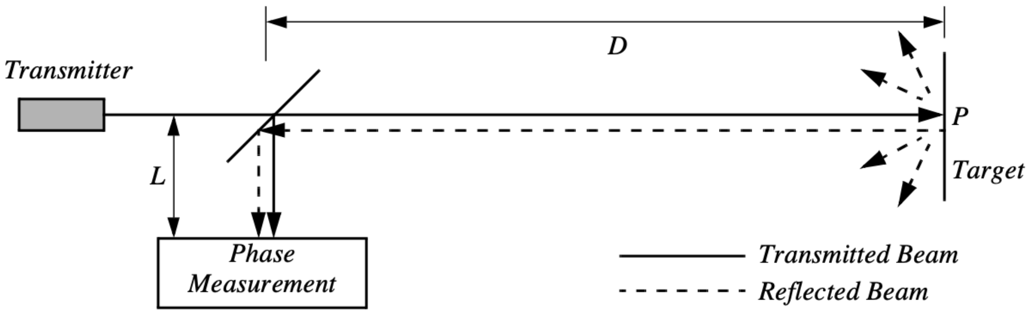


Figure 2.8: Schematic of laser rangefinding by phase-shift measurement.

described in just a little bit. Electronics capable of resolving ps are required in such devices and they are therefore very expensive. A second method is to measure the beat frequency between a frequency modulated continuous wave and its received reflection. Another, even easier method is to measure the phase shift of the reflected light.

Continuous Wave Radar It is a type of radar system where a known stable frequency continuous wave radio energy is transmitted and then received from any reflecting objects. Individual objects can be detected using the Doppler effect, which causes the received signal to have a different frequency from the transmitted signal, allowing it to be detected by filtering out the transmitted frequency.

Doppler-analysis of radar returns can allow the filtering out of slow or non-moving objects, thus offering immunity to interference from large stationary objects and slow-moving clutter. This makes it particularly useful for looking for objects against a background reflector, for instance, allowing a high-flying aircraft to look for aircraft flying at low altitudes against the background of the surface. Because the very strong reflection off the surface can be filtered out, the much smaller reflection from a target can still be seen.

Figure 2.7: A laser range finder used in robotics applications

Phase Shift Measurement Near infrared light, which could be from an Light-Emitting Diode (LED) or a laser, is collimated and transmitted from the transmitter T in **Fig.** 2.8 and hits a point P in the environment.

For surfaces having a roughness greater than the wavelength of the incident light, diffuse reflection will occur, meaning that the light is reflected almost isotropically[12]. The wavelength of the infrared light emitted is 824 nm and so most surfaces with the exception of only highly polished reflecting objects, will be diffuse reflectors. The component of the infrared light which falls within the receiving aperture of the sensor will return almost parallel to the transmitted beam, for distant objects. The sensor transmits 100% amplitude modulated light at a known frequency and measures the phase

[12] Something that is isotropic has the same size or physical properties when it is measured in different directions

shift between the transmitted and reflected signals.

Fig. 2.9 shows how this technique can be used to measure range. The wavelength of the modulating signal obeys the equation $c = f\lambda$ where c is the speed of light and f the modulating frequency.

> For example, f = 5 MHz, the wavelength is λ = 60 m.

The total distance D' covered by the emitted light is:

$$D' = L + 2D = L\frac{\theta}{2\pi}\lambda$$

where D and L are the distances defined in **Fig.** 2.8. The required distance D, between the beam splitter and the target, is therefore given by:

$$D = \frac{\lambda}{4\pi}\theta$$

where θ is the electronically measured phase difference between the transmitted and reflected light beams, and λ the known modulating wavelength. It can be seen that the transmission of a single frequency modulated wave can theoretically result in ambiguous range estimates since

> For example if $\lambda = 60$m, a target at a range of 5 m would give an indistinguishable phase measurement from a target at 65 m, since each phase angle would be 360° apart.

We therefore define an ambiguity interval of λ, but in practice we note that the range of the sensor is much lower than λ due to the attenuation of the signal in air. It can be shown that the confidence in the range (phase estimate) is inversely proportional to the square of the received signal amplitude, directly affecting the sensor's accuracy. Hence dark, distant objects will not produce as good range estimates as close, bright objects.

As with ultrasonic ranging sensors, an important error mode involves coherent reflection of the energy. With light, this will only occur when striking a highly polishes surface. Practically, a AMR may encounter such surfaces in the form of a polished desktop, file cabinet or of course a mirror. Unlike ultrasonic sensors, laser rangefinders cannot detect the presence of optically transparent materials such as glass, and this can be a significant obstacle in environments, for example museums, where glass is commonly used.

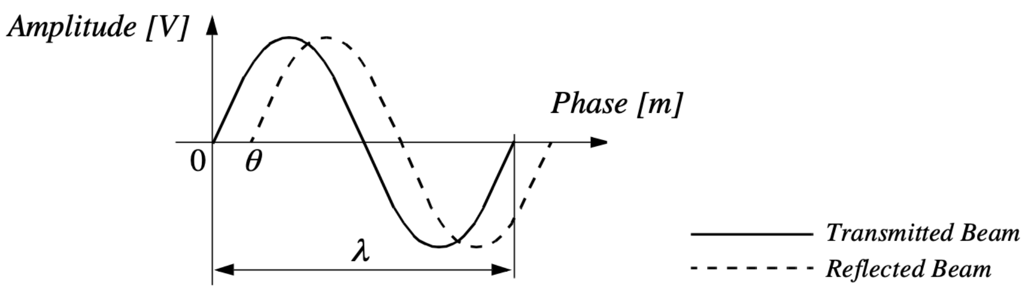


Figure 2.9: Range estimation by measuring the phase shift between transmitted and received signals.

Triangulation-based Active Ranging

Triangulation-based ranging sensors use geometrical properties in their measuring strategy to establish distance readings to objects. The simplest class of triangulation-based rangers are active because they project a known light pattern (e.g., a point, a line or a texture) onto the environment. The reflection of the known pattern is captured by a receiver and, together with known geometric values, the system can use simple triangulation to establish range measurements. If the receiver measures the position of the reflection along a single axis, we call the sensor an optical triangulation sensor in 1D. If the receiver measures the position of the reflection along two orthogonal axes, we call the sensor a structured light sensor.

Optical Triangulation (1D Sensor)

The principle of optical triangulation in 1D is straightforward, as depicted in **Fig.** 2.10. A collimated

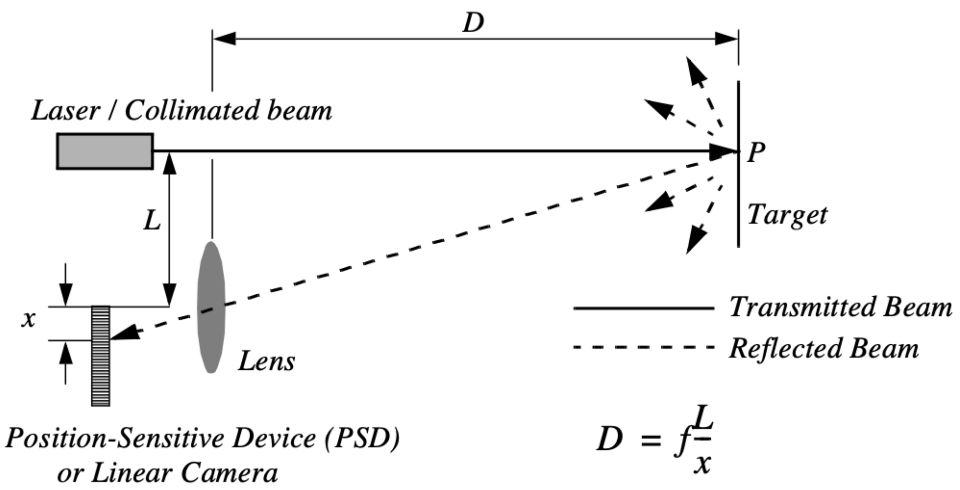


Figure 2.10: Principle of 1D laser triangulation.

beam is transmitted toward the target. The reflected light is collected by a lens and projected onto a position sensitive device[13] or linear camera. Given the geometry of **Fig.** 2.10 the distance D is given by:

$$D = f\frac{L}{x}$$

The distance is proportional to $\frac{1}{x}$, therefore the sensor resolution is best for close objects and becomes worse as distance increases. Sensors based on this principle are used in range sensing up to one or two m, but also in high precision industrial measurements with resolutions far below one ţm. Optical triangulation devices can provide relatively high accuracy with very good resolution for close objects. However, the operating range of such a device is normally fairly limited by geometry. For

[13] A position sensitive device and/or position sensitive detector is an optical position sensor which can measure a position of a light spot in one or two-dimensions on a sensor surface.

example, an off-the-shelf optical triangulation sensor can operate over a distance range of between 8 cm and 80 cm.

It is inexpensive compared to ultrasonic and laser rangefinder sensors.

Although more limited in range than sonar, the optical triangulation sensor has high bandwidth and does not suffer from cross-sensitivities that are more common in the sound domain.

Structured Light (2D Sensor)

If one replaced the linear camera or Position Sensing Device (PSD) of an optical triangulation sensor with a two-dimensional receiver such as a CCD or CMOS camera, then one can recover distance to a large set of points instead of to only one point. The emitter must project a known pattern, or structured light, onto the environment. Many systems exist which either project light textures, which can be seen in **Fig.** 2.12, or emit collimated light by means of a rotating mirror. Yet another popular alternative is to project a laser stripe by turning a laser beam into a plane using a prism. Regardless of how it is created, the projected light has a known structure, and therefore the image taken by the CCD or CMOS receiver can be filtered to identify the pattern's reflection.

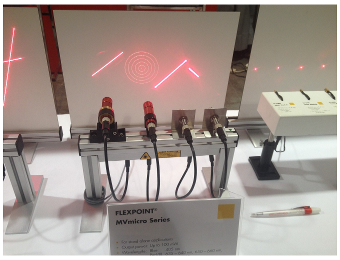


Figure 2.11: Structured light sources on display at the 2014 Machine Vision Show in Boston [36].

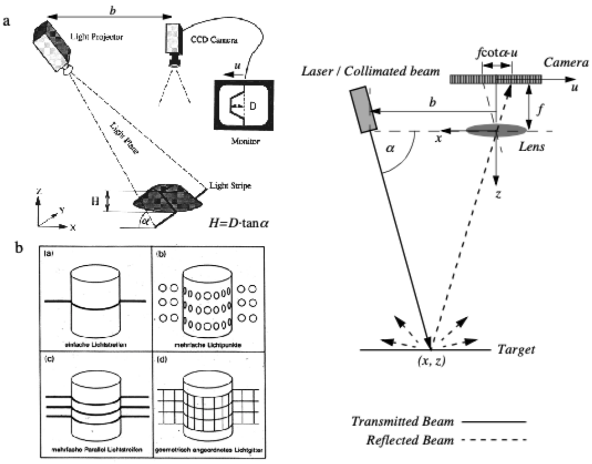


Figure 2.12: a) Principle of active two dimensional triangulation b) Other possible light structures c) One-dimensional schematic of the principle

The problem of recovering depth here far simpler than the problem of passive image analysis.

In passive image analysis, as we discuss later, existing features in the environment must be used to perform correlation, while the present method projects a known pattern upon the environment and thereby avoids the standard correlation problem altogether. Furthermore, the structured light sensor is an active device; so, it will continue to work in dark environments as well as environments in which the objects are featureless[14]. In contrast, stereo vision would fail in such texture-free circumstances. Figure 4.15c shows a one-dimensional active triangulation geometry. We can examine the trade-off in the design of triangulation systems by examining the geometry in figure 4.15c. The measured values in the system are α and u, the distance of the illuminated point from the origin in the imaging sensor.[15] From figure 4.15c, simple geometry shows that:

[14] e.g., uniformly coloured and possess no visible edge.

[15] The imaging sensor here can be a camera or an array of photo diodes of a position sensitive device (e.g. a 2D PSD

$$x = \frac{bu}{f \cot \alpha - u} \qquad \text{and} \qquad z = \frac{bf}{f \cot \alpha - u}.$$

where f is the distance of the lens to the imaging plane. In the limit, the ratio of image resolution to range resolution is defined as the triangulation gain G_{p} and from equation 4.12 is given by:

$$\frac{\partial u}{\partial z} = G_{\text{p}} = \frac{bf}{z^2}$$

This shows that the ranging accuracy, for a given image resolution, is proportional to source/detector separation b and focal length f, and decreases with the square of the range z. In a scanning ranging system, there is an additional effect on the ranging accuracy, caused by the measurement of the projection angle α. From equation 4.12 we see that:

$$\frac{\partial \alpha}{\partial z} = G_{\text{ff}} = \frac{b \sin \alpha^2}{z^2}$$

We can summarise the effects of the parameters on the sensor accuracy as follows:

Baseline Length (b) the smaller b is the more compact the sensor can be.The larger b is the better the range resolution will be. Note also that although these sensors do not suffer from the correspondence problem, the disparity problem still occurs. As the baseline length b is increased, one introduces the chance that, for close objects, the illuminated point(s) may not be in the receiver's field of view.

Detector length and focal length f A larger detector length can provide either a larger field of view or an improved range resolution or partial benefits for both. Increasing the detector length however means a larger sensor head and worse electrical characteristics (increase in random error and reduction of bandwidth). Also, a short focal length gives a large field of view at the expense of accuracy and vice versa.

At one time, laser stripe-based structured light sensors were common on several mobile robot bases as an inexpensive alternative to laser range-finding devices. However, with the in- creasing quality of laser range-finding sensors in the 1990's the structured light system has become relegated largely to vision research rather than applied mobile robotics.

2.2.2 Motion and Speed Sensors

Some sensors directly measure the relative motion between the robot and its environment. Since such motion sensors detect relative motion, so long as an object is moving relative to the robot's

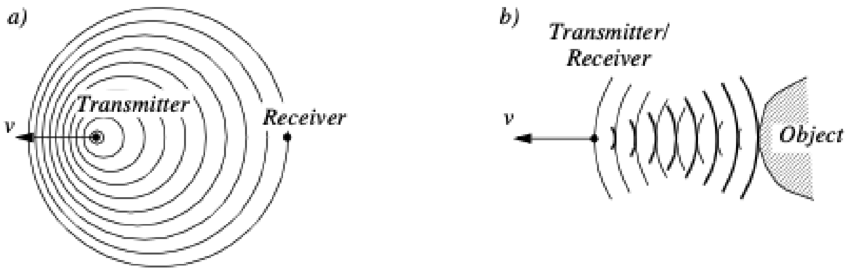
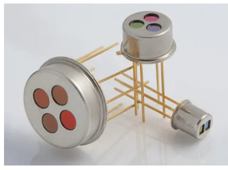


Figure 2.13: Doppler effect between two moving objects (a) or a moving and a stationary object(b)

reference frame, it will be detected and its speed can be estimated. There are a number of sensors that inherently measure some aspect of motion or change.

For example, a pyroelectric[16] sensor detects change in heat.

When someone walks across the sensor's field of view, his motion triggers a change in heat in the sensor's reference frame. In the next subsection, we describe an important type of motion detector based on the Doppler effect. These sensors represent a well-known technology with decades of general applications behind them.



[16] An example of a pyroelectric sensor.

> For fast-moving AMRs such as autonomous highway vehicles and unmanned flying vehicles, Doppler-based motion detectors are the obstacle detection sensor of choice.

Doppler Effect

Anyone who has noticed the change in siren pitch when an ambulance approaches and then passes by is familiar with the Doppler effect.[17]

A transmitter emits an electromagnetic or sound wave with a frequency f t. It is either received by a receiver **Fig.** 2.13(a) or reflected from an object **Fig.** 2.13 (b). The measured frequency f_r at the receiver is a function of the relative speed v between transmitter and receiver according to

$$f_r = f_t \frac{1}{1 + \frac{v}{c}}$$

if the transmitter is moving and

$$f_r = f_t \left(1 + \frac{v}{c}\right)$$

if the receiver is moving. In the case of a reflected wave **Fig.** 2.13 (b) there is a factor of two introduced, since any change x in relative separation affects the round-trip path length by $2x$.

[17] For anyone who needs a bit more information, it is the change in the frequency of a wave in relation to an observer who is moving relative to the source of the wave. The Doppler effect is named after the physicist Christian Doppler, who described the phenomenon in 1842. A common example of Doppler shift is the change of pitch heard when a vehicle sounding a horn approaches and recedes from an observer. Compared to the emitted frequency, the received frequency is higher during the approach, identical at the instant of passing by, and lower during the recession.

In such situations it is generally more convenient to consider the change in frequency Δf, known as the Doppler shift, as opposed to the Doppler frequency notation above.

$$\Delta f = f_t - f_r = \frac{2 f_t \upsilon \cos\theta}{c} \qquad \text{and} \qquad \upsilon = \frac{\Delta f c}{2 f_t \cos\theta}$$

A current application area is both autonomous and manned highway vehicles. Both micro- wave and laser radar systems have been designed for this environment. Both systems have equivalent range, but laser can suffer when visual signals are deteriorated by environmental conditions such as rain, fog, etc. Commercial microwave radar systems are already avail- able for installation on highway trucks. These systems are called VORAD (vehicle on-board radar) and have a total range of approximately 150m. With an accuracy of approximately 97%, these systems report range rate from 0 to 160 km/hr with a resolution of 1 km/ hr. The beam is approximately 4ř wide and 5ř in elevation. One of the key limitations of radar technology is its bandwidth. Existing systems can provide information on multiple targets at approximately 2 Hz.

2.3 Vision Based Sensors

Vision is our most powerful sense. It provides us with an enormous amount of information about the environment and enables rich, intelligent interaction in dynamic environments. It is therefore not at all surprising that a great deal of effort has been devoted to providing machines with sensors which can at least try to mimic the capabilities of the human vision system.

The first step in this process is the creation of sensing devices that capture the same raw information which is the light the human vision system uses. The main topics which will be described are the two (2) current technologies for creating vision sensors:

1. CCD,
2. CMOS.

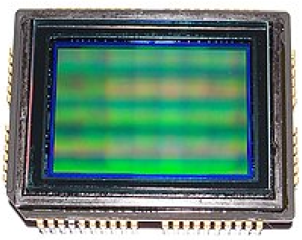


Figure 2.14: Sony ICX493AQA 10.14-megapixel APS-C (23.4 Œ 15.6 mm) CCD from digital camera Sony DSLR-A200 or DSLR-A300, sensor side [37].

Of course, these sensors have specific limitations in performance compared to the human eye, and it is important to understand these limitations. Later sections describe vision-based sensors which are commercially available, similar to the sensors discussed previously, along with their disadvantages and most popular applications.

CCD and CMOS Sensors

When it comes to the marketplace, CCD is the most popular fundamental ingredient for robotic vision systems.[18] The CCD chip, which you can see in **Fig.** 2.14 is an array of light-sensitive picture elements, or pixels, usually with between $2{,}0000 \times 10^4$ and 2 million pixels total.

[18] Willard Boyle and George E. Smith invented the CCD in 1969 at AT&T Bell Labs. Their original idea was to create a memory device. However, with its publication in 1970, other scientists began experimenting with the technology on a range of applications. Astronomers discovered that they could produce high-resolution images of distant objects, because CCDs offered a photo-sensitivity one hundred times greater than film [38].

Each pixel can be thought of as a light-sensitive, discharging capacitor that is 5 to 25 ţm in size. First, the capacitors of all pixels are fully charged, then the integration period begins. As photons of light strike each pixel, the electrons are liberated, which are captured by electric fields and retained at the pixel. Over time, each pixel accumulates a varying level of charge based on the total number of photons that have struck it. After the integration period is complete, the relative charges of all pixels need to be frozen and read.

In a CCD, the reading process is performed at one corner of the CCD chip.[19] The bottom row of pixel charges are transported to this corner and read, then the rows above shift down and the process repeats. This means that each charge must be transported across the chip, and it is critical the value be preserved.

[19] Because the entire array is read through a single amplifier the output can be highly optimised to give very low noise and extremely high dynamic range. CCDs can have over 100 dB dynamic range with less than 2e of noise [38].

This requires specialised control circuitry and custom fabrication techniques to ensure the stability of transported charges.

The photo-diodes used in CCD chips[20] are **NOT** equally sensitive to all frequencies of light. They are sensitive to light between 400 nm and 1000 nm wavelength.[21]

[20] This also includes CMOS as well.

[21] This number range is usually given for easier numbers as both CCD and CMOS have sensitivity values at approximately 350 - 1050 nm.

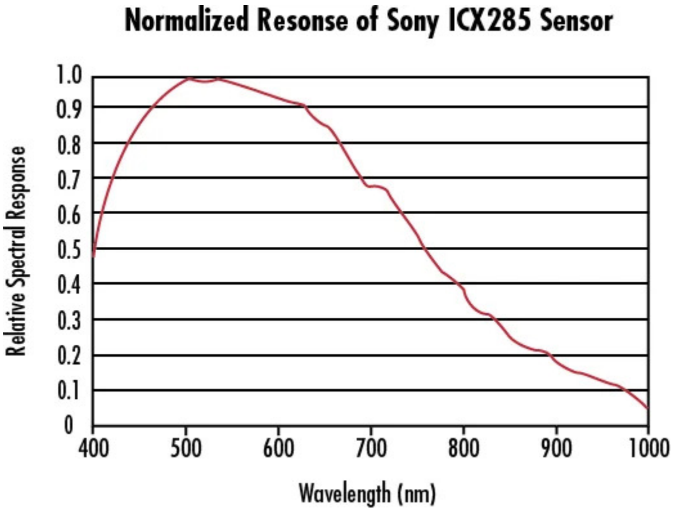
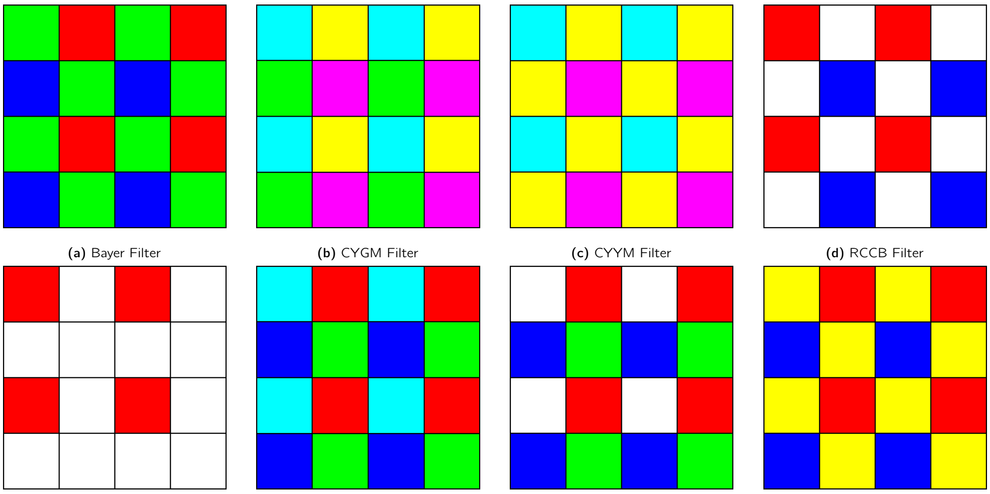


Figure 2.15: Normalized Spectral Response of a Typical Monochrome CCD.

It is important to remember that photodiodes are **less sensitive to the ultraviolet** part of the spectrum and are overly **sensitive to the infrared** portion (e.g. heat) which you can see in **Fig.** 2.15. You can see that the basic light-measuring process is colourless.[22]

[22] It is just measuring the total number of photons that strike each pixel in the integration period regardless whether the light hitting it is blue, red or green.

There are two (2) common approaches for creating color images. If the pixels on the CCD chip are grouped into 2-by-2 sets of four (4), then red, green and blue dyes can be applied to a colour filter so each individual pixel receives only light of just one color.



(a) Bayer Filter **(b)** CYGM Filter **(c)** CYYM Filter **(d)** RCCB Filter **(e)** RCCC Filter **(f)** RGBE Filter **(g)** RGBW Filter **(h)** RYYB Filter

Figure 2.16: Types of colour filter used in commercial and industrial applications

Normally, two (2)pixels measure green while one pixel each measures red and blue light intensity. Of course, this 1-chip color CCD has a geometric resolution disadvantage.

The number of pixels in the system has been effectively cut by a factor of 4, and therefore the image resolution output by the CCD camera will be sacrificed.

The 3-chip color camera avoids these problems by splitting the incoming light into three (3) complete[23] copies. Three separate CCD chips receive the light, with one red, green or blue filter over each entire chip. Thus, in parallel, each chip measures light intensity for just one color, and the camera must combine the CCD chips' outputs to create a joint color image.

[23] Albeit, with lower resolution.

Resolution is preserved in this solution, although the 3-chip color cameras are, as one would expect, significantly more expensive and therefore more rarely used in mobile robotics.

Both 3-chip and single chip color CCD cameras suffer from the fact that photo-diodes are much more sensitive to the near-infrared end of the spectrum. This means that the overall system detects blue light much more poorly than red and green. To compensate, the gain must be increased on the blue channel, and this introduces greater absolute noise on blue[24] than on red and green. It is not uncommon to assume at least 1 - 2 bits of additional noise on the blue channel.

[24] This is generally defined as the amplifier noise.

The CCD camera has several camera parameters that affect its behavior. In some cameras, these parameter values are fixed. In others, the values are constantly changing based on built-in feedback loops. In higher-end cameras, the user can modify the values of these parameters via software embedded into the device. The iris position and shutter speed[25] regulate the amount of light being measured by the camera. The iris is simply a mechanical aperture that constricts incoming light, just as in standard 35mm cameras. Shutter speed regulates the integration period of the chip. In higher-end cameras, the effective shutter speed can be as brief at 1/30,000s and as long as 2s. Camera gain controls the overall amplification of the analog signal, prior to A/D conversion. However, it is very important to understand that, even though the image may appear brighter after setting high gain, the shutter speed and iris may not have changed at all. Thus gain merely amplifies the signal, and amplifies along with the signal all of the associated noise and error. Although useful in applications where imaging is done for human consumption (e.g. photography, television), gain is of little value to a mobile roboticist.

[25] It's the speed at which the shutter of the camera closes. A fast shutter speed creates a shorter exposure - the amount of light the camera takes in - and a slow shutter speed gives a longer exposure.

In colour cameras, an additional control exists for white balance. Depending on the source of illumination in a scene[26] the relative measurements of red, green and blue light which combine to define pure white light will change dramatically which can be seen in **Fig.** 2.17 which can also be adjusted with algorithms [39]. The human eyes compensate for all such effects in ways that are not fully understood, however, the camera can demonstrate glaring inconsistencies in which the same table looks blue in one image, taken during the night, and yellow in another image, taken during the day. White balance controls enable the user to change the relative gain for red, green and blue in order to maintain more consistent color definitions in varying contexts.

[26] For example this could be fluorescent lamps, incandescent lamps, sunlight, underwater filtered light, etc.

The key disadvantages of CCD cameras are primarily in the areas of inconstancy and **dynamic range**.

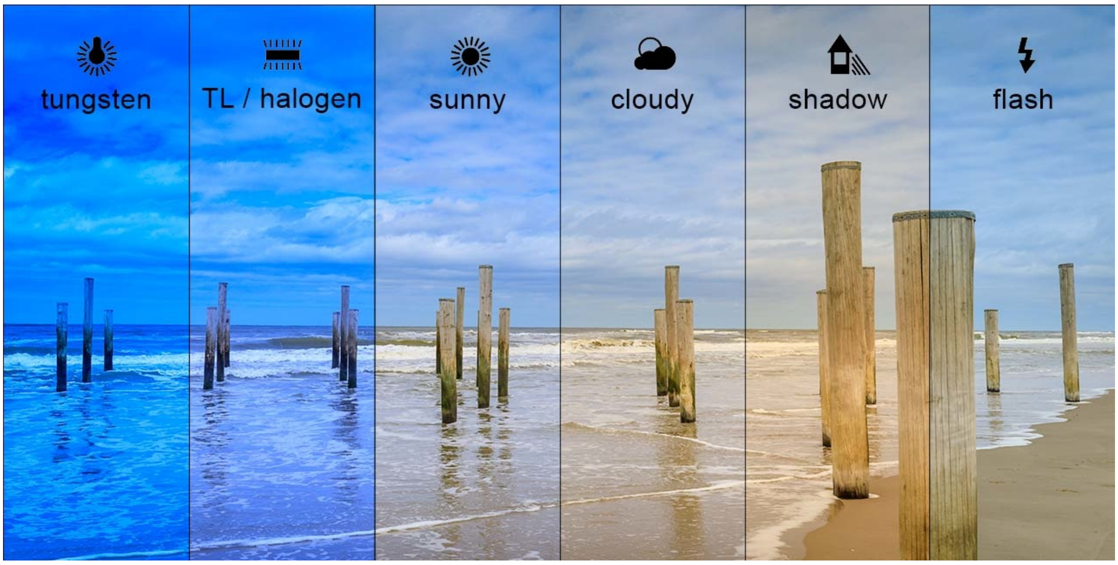


Figure 2.17: Example of white balance. Here the same scene is emulated to be shot under different light conditions [40].

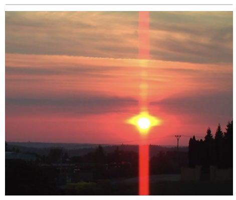
Information: Dynamic Range

Dynamic range in photography describes the ratio between the maximum and minimum measurable light intensities (white and black, respectively). In the real world, one never encounters true white or black - only varying degrees of light source intensity and subject reflectivity. Therefore the concept of dynamic range becomes more complicated, and depends on whether you are describing a capture device (such as a camera or scanner), a display device (such as a print or computer display), or the subject itself.

As mentioned above, a number of parameters can change the brightness and colours with which a camera creates its image.

Manipulating these parameters in a way to provide consistency over time and over environments, for example ensuring a green shirt always looks green, and something dark grey is always dark grey, remains an open problem [41].

The second type of disadvantages relates to the behavior of a CCD chip in environments with extreme illumination. In cases of very low illumination, each pixel will receive only a small number of photons. The longest possible shutter speed and camera optics (i.e. pixel size, chip size, lens focal length and diameter) will determine the minimum level of light for which the signal is stronger than random error noise. In cases of very high illumination, a pixel fills its well with free electrons and, as the well reaches its limit, the probability of trapping additional electrons falls and therefore the linearity between incoming light and electrons in the well degrades. This is termed saturation[27] and can indicate the existence of a further problem related to cross-sensitivity [43]. When a well has reached its limit, then additional light within the remainder of the integration period may cause further charge to leak into neighbouring pixels, causing them to report incorrect values or even reach secondary saturation. This effect, called blooming, means that individual pixel values are **NOT** truly independent. The camera parameters may be adjusted for an environment with a particular light



[27] Example of blooming caused by saturation of a sensor pixel. The sun is so bright in the image that there is blooming on the sun itself, leaking into the surrounding pixels, and a vertical smear across the whole image [42].

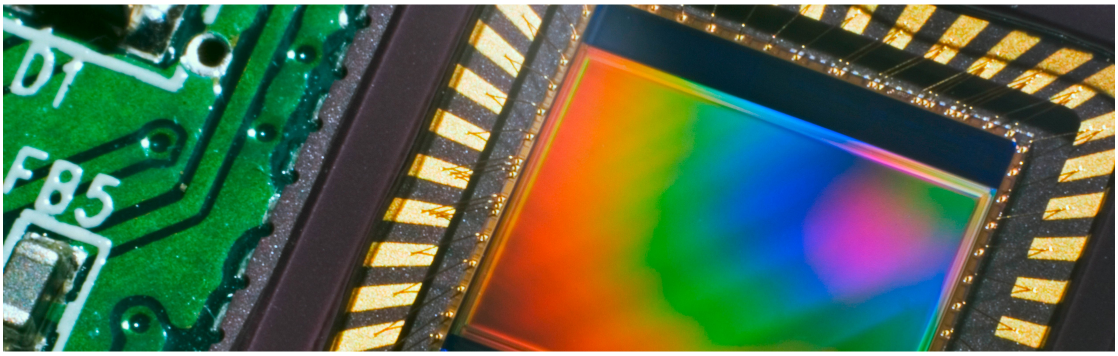


Figure 2.18: A close-up view of a CMOS sensor and its circuitry [44].

level, but the problem remains that the dynamic range of a camera is limited by the well capacity of the individual pixels.

> For example, a high quality CCD may have pixels that can hold 4,0000 $\times$ 10^4 electrons. The noise level for reading the well may be 11 electrons, and therefore the dynamic range will be 40,000:11, or 3,600:1, which is 35 dB.

2.3.1 CMOS Technology

The Complementary Metal Oxide Semiconductor (CMOS) chip is a significant departure from the CCD. Similar to CCD, it too has an array of pixels, but located alongside each pixel are several transistors specific to that pixel. Just as in CCD chips, all of the pixels accumulate charge during the integration period. During the data collection step, the CMOS takes a new approach:

> The pixel-specific circuitry next to every pixel measures and amplifies the pixel's signal, all in parallel for every pixel in the array.

Using more traditional traces from general semiconductor chips, the resulting pixel values are all carried to their destinations. CMOS has a number of advantages over CCD technologies. First and foremost, there is no need for the specialized clock drivers and circuitry required in the CCD to transfer each pixel's clock down all of the array columns and across all of its rows.[28]

> This also means that specialized semiconductor manufacturing processes are not required to create CMOS chips.

Therefore, the same production lines that create microchips can create inexpensive CMOS chips as well. The CMOS chip is so much simpler that it consumes significantly less power, it operates with a power consumption a tenth the power consumption of a CCD chip [46].

> In a AMR, power is a scarce resource and therefore this is an important advantage.

On the other hand, the CMOS chip also faces several disadvantages.

[28]-CAM80CUNX is an 8MP Ultra-lowlight MIPI CSI-2 camera capable of streaming 4K @ 44 fps. This 8MP camera is based on SONY STARVIS IMX415 CMOS image sensor [45]

- Most importantly, the circuitry next to each pixel consumes valuable real estate on the face of the light-detecting array. Many photons hit the transistors rather than the photodiode, making the CMOS chip significantly less sensitive than an equivalent CCD chip.
- CMOS, compared to CCD is still finding ground in the marketplace, and as a result, the best resolution that one can purchase in CMOS format continues to be far inferior to the best CCD chips available.
- CMOS sensors have a lower dynamic range,
- CMOS sensors have higher levels of noise.

Compared to the human eye, these chips all have worse performance, cross-sensitivity and a limited dynamic range. As a result, vision sensors today continue to be fragile. Only over time, as the underlying performance of imaging chips improves, will significantly more robust vision-based sensors for AMRs be available.

Information: Shot Noise

Shot noise or Poisson noise is a type of noise which can be modeled by a Poisson process.

In electronics shot noise originates from the discrete nature of electric charge. Shot noise also occurs in photon counting in optical devices, where shot noise is associated with the particle nature of light.

Figure 2.19: Photon noise simulation. Number of photons per pixel increases from left to right and from upper row to bottom row [47].

2.3.2 Visual Ranging Sensors

Range sensing is extremely important in AMR as it is a basic input for successful obstacle avoidance. As we have seen earlier, a number of sensors are popular in robotics specifically for their ability to recover depth estimates:

ultrasonic, laser rangefinder, optical rangefinder, etc.

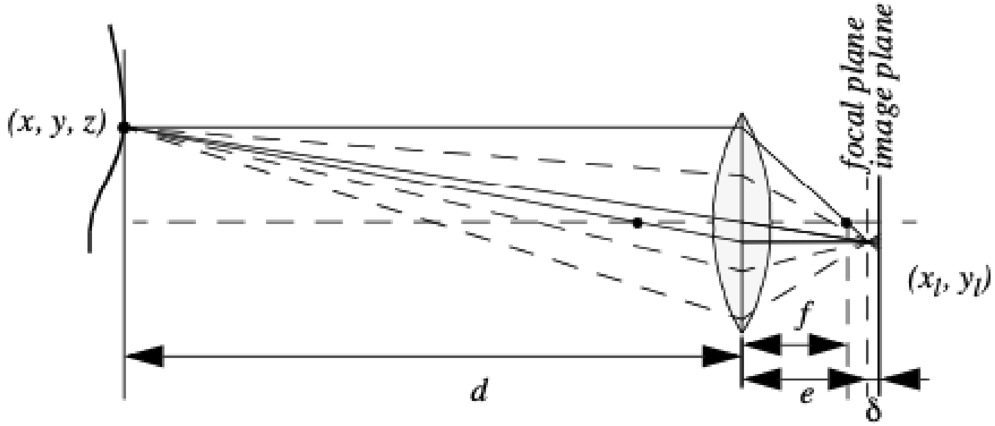


Figure 2.20: Depiction of the camera optics and its impact on the image. To get a sharp image, the image plane must coincide with the focal plane. Otherwise the image of the point (x, y, z) will be blurred in the image as can be seen in the drawing above.

It is natural to attempt to implement ranging functionality using vision chips as well. However, a fundamental problem with visual images makes rangefinding relatively difficult.

Any vision chip collapses the three-dimensional world into a two-dimensional image plane, thereby losing depth information. If one can make strong assumptions regarding the size of objects in the world, or their particular colour and reflectance, then one can directly interpret the appearance of the two-dimensional image to recover depth. But such assumptions are rarely possible in real-world AMR applications.

Without such assumptions, a single picture does not provide enough information to recover spatial information.

The general solution is to recover depth by looking at several images of the scene to gain more information, which will be hopefully enough to at least partially recover depth. The images used must be different, so that taken together they provide additional information. They could differ in viewpoint, which would allow the use of stereo or motion algorithms.

An alternative is to create different images, not by changing the viewpoint, but by changing the camera geometry, such as the focus position or lens iris. This is the fundamental idea behind depth from focus and depth from defocus techniques. We will now look into the general approach to the depth from focus techniques as it presents a straightforward and efficient way to create a vision-based range sensor.

2.3.3 Depth from Focus

The depth from focus class of techniques relies on the fact that image properties not only change as a function of the scene, but also as a function of the camera parameters. The relationship between camera parameters and image properties is depicted in **Fig.** 2.20. The fundamental formula governing image formation relates the distance of the object from the lens, d in **Fig.** 2.20, to the distance e from the lens to the focal point, based on the focal length f of the lens:

$$\frac{1}{f} = \frac{1}{d} + \frac{1}{e}$$

If the image plane is located at distance e from the lens, then for the specific object voxel[29] depicted, all light will be focused at a single point on the image plane and the object voxel will be focused. However, when the image plane is **NOT** at e, as is seen in **Fig.** 2.20, then the light from the object voxel will be cast on the image plane as a blur circle. To a first approximation, the light is homogeneously distributed throughout this blur circle, and the radius R of the circle can be characterized according to the equation:

$$R = \frac{L\delta}{2e}$$

where L is the diameter of the lens or aperture and δ is the displacement of the image plan from the focal point.

Given these formulae, several basic optical effects are clear.

> For example, if the aperture[30] or lens is reduced to a point, as in a pin-hole camera, then the radius of the blur circle approaches zero.

This is consistent with the fact that decreasing the iris aperture opening causes the depth of field to increase until all objects are in focus. Of course, the disadvantage of doing so is that we are allowing less light to form the image on the image plane and so this is practical only in bright circumstances The second property to be deduced from these optics equations relates to the sensitivity of blurring as a function of the distance from the lens to the object.

Suppose the image plane is at a fixed distance 1.2 from a lens with diameter L = 0.2 and focal length f = 0.5. We can see from Equation (4.20) that the size of the blur circle R changes proportionally with the image plane displacement . If the object is at distance d = 1, then from Equation (4.19) we can compute e=1 and therefore = 0.2. Increase the object distance to d = 2 and as a result = 0.533. Using Equation (4.20) in each case we can compute R = 0.02 R = 0.08 respectively. This demonstrates high sensitivity for defocusing when the object is close to the lens. In contrast suppose

Figure 2.21: Three images of the same scene taken with a camera at three different focusing positions. Note the significant change in texture sharpness between the near surface and far surface [48].

[29] A three-dimensional counterpart to a pixel.

[30] The aperture is the opening in the lens that allows light to enter the camera and onto the sensor or film.

the object is at d = 10. In this case we compute e = 0.526. But if the object is again moved one unit, to d = 11, then we compute e = 0.524. Then resulting blur circles are R = 0.117 and R = 0.129, far less than the quadrupling in R when the obstacle is 1/10 the distance from the lens. This analysis demonstrates the fundamental limitation of depth from focus techniques: they lose sensitivity as objects move further away (given a fixed focal length). Interestingly, this limitation will turn out to apply to virtually all visual ranging techniques, including depth from stereo and depth from motion. Nevertheless, camera optics can be customised for the depth range of the intended application. For example, a "zoom" lens with a very large focal length f will enable range resolution at significant distances, of course at the expense of field of view. Similarly, a large lens diameter, coupled with a very fast shutter speed, will lead to larger, more detectable blur circles. Given the physical effects summarised by the above equations, one can imagine a visual ranging sensor that makes use of multiple images in which camera optics are varied (e.g. image plane displacement) and the same scene is captured (see Fig. 4.20). In fact this approach is not a new invention. The human visual system uses an abundance of cues and techniques, and one system demonstrated in humans is depth from focus. Humans vary the focal length of their lens continuously at a rate of about 2 Hz. Such approaches, in which the lens optics are actively searched in order to maximise focus, are technically called depth from focus. In contrast, depth from defocus means that depth is recovered using a series of images that have been taken with different camera geometries. Depth from focus methods are one of the simplest visual ranging techniques. To determine the range to an object, the sensor simply moves the image plane (via focusing) until maximizing the sharpness of the object. When the sharpness is maximised, the corresponding position of the image plane directly reports range. Some autofocus cameras and virtually all autofocus video cameras use this technique. Of course, a method is required for measuring the sharpness of an image or an object within the image. The most common techniques are approximate measurements of the sub-image gradient:

$$\text{sharpness}_1 = \sum_{x,y} \left| I(x, y) - I(x-1, y) \right| \tag{2.3}$$

$$\text{sharpness}_2 = \sum_{x,y} \left(I(x, y) - I(x-2, y-2) \right)^2 \tag{2.4}$$

A significant advantage of the horizontal sum of differences technique (Equation (4.21)) is that the calculation can be implemented in analog circuitry using just a rectifier, a low-pass filter and a high-pass filter. This is a common approach in commercial cameras and video recorders. Such systems will be sensitive to contrast along one particular axis, although in practical terms this is rarely an issue. However depth from focus is an active search method and will be slow because it takes time to change the focusing parameters of the camera, using for example a servo-controlled focusing ring. For this reason this method has not been applied to AMRs. A variation of the depth from focus technique has been applied to a AMR, demonstrating obstacle avoidance in a variety of environments as well as avoidance of concave obstacles such as steps and ledges [95]. This robot uses three monochrome cameras placed as close together as possible with different, fixed lens focus positions (Fig. 4.21).

Several times each second, all three frame-synchronised cameras simultaneously capture three images of the same scene. The images are each divided into five columns and three rows, or 15 subregions. The approximate sharpness of each region is computed using a variation of Equation (4.22), leading to a total of 45 sharpness values. Note that Equation 22 calculates sharpness along diagonals but skips one row. This is due to a subtle but important issue. Many cameras produce images in interlaced mode. This means that the odd rows are captured first, then afterwards the even rows are captured. When such a camera is used in dynamic environments, for example on a moving robot, then adjacent rows show the dynamic scene at two different time points, differing by up to 1/30 seconds. The result is an artificial blurring due to motion and not optical defocus. By comparing only even-number rows we avoid this interlacing side effect.

Recall that the three images are each taken with a camera using a different focus position. Based on the focusing position, we call each image close, medium or far. A 5x3 coarse depth map of the scene is constructed quickly by simply comparing the sharpness values of each three corresponding regions. Thus, the depth map assigns only two bits of depth information to each region using the values close, medium and far. The critical step is to adjust the focus positions of all three cameras so that flat ground in front of the obstacle results in medium readings in one row of the depth map. Then, unexpected readings of either close or far will indicate convex and concave obstacles respectively, enabling basic obstacle avoidance in the vicinity of objects on the ground as well as drop-offs into the ground. Although sufficient for obstacle avoidance, the above depth from focus algorithm presents unsatisfyingly coarse range information. The alternative is depth from defocus, the most desirable of the focus-based vision techniques. Depth from defocus methods take as input two or more images of the same scene, taken with different, known camera geometry. Given the images and the camera geometry settings, the goal is to recover the depth information of the three-dimensional scene represented by the images. We begin by deriving the relationship between the actual scene properties (irradiance and depth), camera geometry settings and the image g that is formed at the image plane. The focused image f(x,y) of a scene is defined as follows. Consider a pinhole aperture (L=0) in lieu of the lens. For every point p at position (x,y) on the image plane, draw a line through the pinhole aperture to the corresponding, visible point P in the actual scene. We define f(x,y) as the irradiance (or light intensity) at p due to the light from P. Intuitively, f(x,y) rep- resents the intensity image of the scene perfectly in focus

2.4 Feature Extraction

An AMR must be able to determine its relationship to the environment by making measurements with its sensors and then using those measured signals. A wide variety of sensing technologies are available, as we discussed previously. But every sensor we have presented is imperfect:

> measurements always have error and, therefore, un- certainty associated with them.

Therefore, sensor inputs must be used in a way that enables the robot to interact with its environment successfully in spite of measurement uncertainty. There are two (2) strategies for using uncertain sensor input to guide the robot's behavior. One strategy is to use each sensor measurement as a raw and individual value. Such raw sensor values could for example be tied directly to robot behavior, whereby the robot's actions are a function of its sensor inputs. Alternatively, the raw sensors values could be used to update an intermediate model, with the robot's actions being triggered as a function of this model rather than the individual sensor measurements.

The second strategy is to extract information from one or more sensor readings first, gener- ating a higher-level percept that can then be used to inform the robot's model and perhaps the robot's actions directly. We call this process feature extraction, and it is this next, op- tional step in the perceptual interpretation pipeline (Fig. 4.34) that we will now discuss.

In practical terms, mobile robots do not necessarily use feature extraction and scene inter- pretation for every activity. Instead, robots will interpret sensors to varying degrees depend- ing on each specific functionality. For example, in order to guarantee emergency stops in the face of immediate obstacles, the robot may make direct use of raw forward-facing range readings to stop its drive motors. For local obstacle avoidance, raw ranging sensor strikes may be combined in an occupancy grid model, enabling smooth avoidance of obstacles meters away. For map-building and precise navigation, the range sensor values and even vision sensor measurements may pass through the complete perceptual pipeline, being sub- jected to feature extraction followed by scene interpretation to minimize the impact of indi- vidual sensor uncertainty on the robustness of the robot's map-making and navigation skills. The pattern that thus emerges is that, as one moves into more sophisticated, long-term per- ceptual tasks, the feature extraction and scene interpretation aspects of the perceptual pipe- line become essential.

2.4.1 Defining Feature

Features are recognizable structures of elements in the environment. They usually can be extracted from measurements and mathematically described. Good features are always perceivable and easily detectable from the environment. We distinguish between low-level fea- tures (geometric primitives) like lines, circles or polygons and high-level features (objects) such as edges, doors, tables or a trash can. At one extreme, raw sensor data provides a large volume of data, but with low distinctiveness of each individual quantum of data. Making use of raw data has the potential advantage that every bit of information is fully used, and thus there is a high conservation of information. Low level features are abstractions of raw data, and as such provide a lower volume of data while increasing the distinctiveness of each fea- ture. The hope, when one incorporates low level features, is that the features are filtering out poor or useless data, but of course it is also likely that some valid information will be lost as a result of the feature extraction process. High level features provide maximum ab- straction from the raw data, thereby reducing the volume of data as much as possible while providing highly distinctive resulting features. Once again, the abstraction process has the risk of filtering away important information, potentially lowering data utilization.

Although features must have some spatial locality, their geometric extent can range widely. For example, a corner feature inhabits a specific coordinate location in the geometric world. In contract, a visual "fingerprint" identifying a specific room in an office building applies to the entire room, but has a location that is spatially limited to the one, particular room. In mobile robotics, features play an especially important role in the creation of environmen- tal models. They enable more compact and robust descriptions of the environment, helping a mobile robot during both map-building and localization. When designing a mobile robot, a critical decision revolves around choosing the appropriate features for the robot to use. A number of factors are essential to this decision:

Target Environment For geometric features to be useful, the target geometries must be readily detected in the actual environment. For example, line features are extremely useful in office building environments due to the abundance of straight walls segments while the same feature is virtually useless when navigating Mars.

Available Sensors Obviously the specific sensors and sensor uncertainty of the robot im- pacts the appropriateness of various features. Armed with a laser rangefinder, a robot is well qualified to use geometrically detailed features such as corner features due to the high qual- ity angular and depth resolution of the laser scanner. In contrast, a sonar-equipped robot may not have the appropriate tools for corner feature extraction.

Computational Power Vision-based feature extraction can effect a significant computa- tional cost, particularly in robots where the vision sensor processing is performed by one of the robot's main processors.

Environment representation Feature extraction is an important step toward scene inter- pretation, and by this token the features extracted must provide information that is consonant with the representation used for the environment model. For example, non-geometric vi- sion-based features are of little value in purely geometric environment models but can be of great value in topological models of the environment. Figure 4.35 shows the application of two different representations to the task of modeling an office building hallway. Each ap- proach has advantages and disadvantages, but extraction of line and corner features has much more relevance to the representation on the left. Refer to Chapter 5, Section 5.5 for a close look at map representations and their relative tradeoffs. In the following two sections, we present specific feature extraction techniques based on the two most popular sensing modalitites of mobile robotics: range sensing and visual appear- ance-based sensing.

2.4.2 Using Range Data

Most of today's features extracted from ranging sensors are geometric primitives such as line segments or circles. The main reason for this is that for most other geometric primitives the parametric description of the features becomes too complex and no closed form solution exists. Here we will describe line extraction in detail, demonstrating how the uncertainty models presented above can be applied to the problem of combining multiple sensor mea- surements. Afterwards, we briefly present another very successful feature for indoor mobile robots, the corner feature, and demonstrate how these features can be combined in a single representation.

Line Extraction

Geometric feature extraction is usually the process of comparing and matching measured sensor data against a predefined description, or template, of the expect feature. Usually, the system is overdetermined in that the number of sensor measurements exceeds the number of feature parameters to be estimated. Since the sensor measurements all have some error, there is no perfectly consistent solution and, instead, the problem is one of optimization. One can, for example, extract the feature that minimizes the discrepancy with all sensor measurements used (e.g. least squares estimation). In this section we present an optimization-based solution to the problem of extracting a line feature from a set of uncertain sensor measurements. For greater detail than is presented below, refer to [19], pp. 15 and 221.

Probabilistic Line Extraction

4.36. There is uncertainty associated with each of the noisy range sensor measurements, and so there is no single line that passes through the set. Instead, we wish to select the best pos- sible match, given some optimization criterion. More formally, suppose n ranging measurement points in polar coordinates x = (,) iii are produced by the robot's sensors. We know that there is uncertainty associated with each measurement, and so we can model each measurement using two random variables X = (P , Q) . In this analysis we assume that uncertainty with respect to the actual value iii of P and Q are independent. Based on Equation (4.56) we can state this formally: Furthermore, we will assume that each random variable is subject to a Gaussian probability density curve, with a mean at the true value and with some specified variance: Given some measurement point (,) , we can calculate the corresponding Euclidean co- ordinatesasx = cos andy = sin. Iftherewerenoerror,wewouldwanttofinda line for which all measurements lie on that line: Of course there is measurement error, and so this quantity will not be zero. When it is non-zero, this is a measure of the error between the measurement point (,) and the line, spe- cifically in terms of the minimum orthogonal distance between the point and the line. It is always important to understand how the error that shall be minimized is being measured. For example a number of line extraction techniques do not minimize this orthogonal point- line distance, but instead the distance parallel to the y-axis between the point and the line. A good illustration of the variety of optimization criteria is available in [18] where several algorithms for fitting circles and ellipses are presented which minimize algebraic and geo- metric distances. For each specific (,) , we can write the orthogonal distance d between (,) and iiiii the line as:

Part II

Localisation and Mapping

If you don't know where you want to go, then it doesn't matter which path you take.

(Lewis Carroll, Alice in Wonderland)

Part III

GNU/Linux Operating System

Hello everybody out there using MINIX.

I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu) for 386(486) AT clones. This has been brewing since April, and is starting to get ready. I'd like any feedback on things people like/dislike in minix, as my OS resembles it somewhat (same physical layout of the file-system (due to practical reasons) among other things).

I've currently ported bash (1.08) and GCC (1.40), and things seem to work. This implies that I'll get something practical within a few months, and I'd like to know what features most people would want. Any suggestions are welcome, but I won't promise I'll implement them :-)
- Linus

PS. Yes - it's free of any MINIX code, and it has a multi-threaded fs. It is NOT portable (uses 386 task switching etc.), and it probably never will support anything other than AT-harddisks, as that's all I have :-(

(Linus Torvalds sending an email to comp.os.inix (Edited))

Part IV

Robot Operating System

1: A robot may not injure a human being or, through inaction, allow a human being to come to harm;

2: A robot must obey the orders given it by human beings except where such orders would conflict with the First Law;

3: A robot must protect its own existence as long as such protection does not conflict with the First or Second Law;

The Zeroth Law: A robot may not harm humanity, or, by inaction, allow humanity to come to harm.

(Three Laws of Robotics, Isaac Asimov, I, Robot)

Glossary

AMR Autonomous Mobile Robotics. 5, 27–30, 32–37, 39–43, 45, 49, 55–57, 59, 61

CCD Charge Coupled Device. vi, 28, 32, 47, 51–56

CMOS Complimentary MOS. vi, 47, 51, 52, 55, 56

DoF Degrees of Freedom. v, 7, 10–12

GPS Global Positioning System. 38

LED Light-Emitting Diode. 44

LIDAR Light Detection and Ranging. 43

PSD Position Sensing Device. 47, 48

ToF Time-of-Flight. 40, 41, 43

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